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CMPU Aged Order Analysis

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Date: November 4, 2026

1 Executive summary

For CMPU (Customer Must Pickup and Upload) vet diet orders; the customer is required to pick up the their prescriptions and upload them to Chewy. This order group suffers from high cancellation rates; the top cancellation reason begin “aged order”; an automatic cancellation that happens if the order has not been shipped or otherwise been canceled by the customer after ten days from order creation. Aged order cancellation is followed by explicit cancellations by the customer (“customer initiated cancellation” and “customer request”). Unfortunately, cancellation reasons do not provide actionable root causes and require further investigation.

This report focuses on aged order cancellations. Among CMPU vet diet orders shipped or canceled between May 1, 2026 and July 31, 2026, 38,314 orders were canceled due to aging, with recorded line net sales of $3,470,380.19 as an opportunity baseline. These amounts are 9.88% of the 387,707 orders and 9.69% of the $35,819,402.26 in line net sales in the no-recorded-cancellation-reason group, respectively (Table 1). That comparison group is not a verified shipped-only population, and the aged-order amount is not an estimate of recoverable revenue. Customer initiated and customer requested cancellations have a combined recorded line-sales baseline of $3,988,013.32.

Within seven days after an aged cancellation, 2,316 of 30,382 customers (7.62%) place a qualifying non-cancelled reorder for the same pet and part number (Table 2). Those reorders have $221,122.35 in recorded net sales, equivalent to 6.37% of the $3,470,380.19 aged-order sales baseline. At 28 days, recorded follow-up net sales are $685,508.89, or 19.75% of that same baseline. These are observed follow-up sales, not measured incremental recovery. By 70 days, 7,944 of 30,382 customers (26.15%) reorder the same product for the same pet. Among the 11,550 customers whose first observed customer-upload vet-diet order aged, 833 (7.21%) reorder within seven days and 2,509 (21.72%) within 70 days. Among the 5,097 customers whose first observed vet-diet order aged, 800 (15.70%) reorder within 70 days, using the same pet/product endpoint (Table 2). For any qualifying Chewy reorder within 70 days, the corresponding counts are 18,564/30,382 (61.10%), 6,712/11,550 (58.11%) and 2,591/5,097 (50.83%), respectively (Table 5). The reorder percentage is lowest in the new to vet diet subgroup.

All of the customers seem to have contacted Chewy, most of them at the creation of the order (Table 7). Some of them have contacted Chewy on multiple days (Table 8). People who contacted closer to the aged order cancellation date have a higher reorder rate (Table 10).

In the August–October prescription study, 777 of 26,443 aged first-order lines have a matched upload (2.94%); 25,666 (97.06%) have none in the retained source window (Table 14). This measures matched evidence, not proof that a customer never uploaded. Compared to the overall upload coverage of 31.18% , this low coverage suggests that there is a root cause for aged orders leading to lower than average upload rates. Of the 782 distinct matched upload entities, 52 are currently ACCEPTED (6.65%); entity acceptance is distinct from prescription approval. These snapshot counts allow uploads and state changes after cancellation. Of those prescriptions with uploads, the time between order creation and upload does not significantly differ from the rest of the study group (Figure 2). The accepted aged-order entities are mostly accepted late; after 10 days (Figure 4). Approval after acceptance is not a significant contributor to the age of the order most of the time; it takes less than a minute after acceptance (95% of the time). (Figure 5). Finally, Chewy can expect that if the customer does not upload a prescription in 3 days after order creation; there is more about 50% chance that the order will be canceled as an aged order (Figure 6).

So what can be done to reduce aged orders? One suggestion to increase the upload rate (communication with Courtney Gleede) is to send reminder emails to customer. At this point, Chewy does not contact the at risk orders’ customers. There is no Chewy relevant out-bound contact data for a comparative/causal analysis. The only contact happens when the customer is notified that their order has been canceled. Assuming that people who receive an email before cancellation react similarly to the cohort that reorders within 7 days after cancellation, we expect that out of 30K people considered in the opportunity study with aged cancellations, 2.3K would react before order cancellation corresponding to 221K sales. The best effort guess is that Chewy prevents people from skipping one reorder for this cohort for an additional gain of 221K net sales. The upper bound on the best effort estimate is 390K corresponding to the 14 day reorder window cohort. For scale, that cohort’s $386,124.68 in observed follow-up net sales is 1.08% of the $35,819,402.26 no-recorded-cancellation-reason sales baseline in Table 1; this ratio is not a measured incremental sales increase from reminders.

The effect of reminder emails on long term retention is an open question. Picking up and uploading a prescription is still an onus with or without a getting a reminder.

2 What are the next steps?

1. The root causes of aged orders and the other customer initiated/requested cancellations are not known. We know that highly likely they did not upload their prescriptions but why they have not uploaded is still unknown. In order to get more

insight into the root causes; Chewy should

(a) design and apply a survey to customers with canceled orders (aged or customer requested/initiated) either in person or via AI (AI calls or emails) to understand cancellation root causes and how Chewy can help them; such as a prescription pick-up service; focusing on first time vet diet customers, as their retention is the worst.

(b) analyze the customer contact logs for plausible insights into order cancellation root causes.

2. design and execute a controlled experiment where Chewy sends reminder emails to a group of customers who have not yet uploaded their prescriptions 3 days after order creation to estimate the uplift from reminder emails.

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3.1 What is the financial impact of CMPU aged orders?

Table 1 reports distinct customer and order counts, qualifying order-line counts and recorded net sales by cancellation reason and line-item status for orders shipped or cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC (`[2026-05-01, 2026-08-01)`, with an exclusive end date). The population requires vet-diet lines matched by prescription ID and numeric customer ID to vet-diet prescriptions with `PRESCRIPTION_SOURCE = CMPU` and `APPROVAL_CHANNEL = CUSTOMER_UPLOAD`.

3.1.1 What is the estimated net sales lost from CMPU aged orders?

3.1.1.1 How do customer counts, order counts, and estimated net sales vary by cancellation reason and line-item status?

Financial opportunity baseline by cancellation reason and line-item status: customers, orders, qualifying order lines and net sales for CMPU vet-diet customer-upload orders shipped or cancelled May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 1: Financial opportunity baseline by cancellation reason and line-item status: customers, orders, qualifying order lines and net sales for CMPU vet-diet customer-upload orders shipped or cancelled May 1, 2026 through July 31, 2026, inclusive, UTC.

Cancellation reason	Line-item status	Customers	Orders	Order lines	Estimated net sales
[No recorded cancellation reason] (actual net sales)	D	172,621	387,273	432,587	35,782,681.49
[No recorded cancellation reason]	X	432	434	464	36,720.77
Aged order	X	30,382	38,314	41,713	3,470,380.19
Customer initiated cancellation	X	29,455	32,939	37,959	2,945,818.13
Customer request	X	11,382	11,948	13,662	1,042,195.19
Duplicate order	X	161	165	188	14,763.64
Fraud	X	36	38	45	4,339.97
Inventory issue	X	5	5	5	583.24
No response	X	1,035	1,229	1,329	111,509.37
Other	X	784	809	940	71,158.59
Payment declined	X	4,242	5,533	6,067	523,931.07
Payment issue	X	46	47	57	4,207.82
Pharmacy rejection	X	278	310	351	29,055.56
Prescription issue	X	5,051	5,329	5,822	480,281.17
Restricted availability	X	1	1	1	25.64
TOTAL	All	225,434	484,374	541,190	44,517,651.84

Notes: Includes all qualifying orders, including subsequent orders; shipment or cancellation records must fall in the reporting window. There is no order-placement or prescription-creation restriction. This differs from the prescription-creation cohort and aged-cancellation cutoff used in Table 11. Counts are distinct customers, orders and physical order lines. Customers may appear in multiple reason/status rows; TOTAL counts each customer once. Sales include only qualifying lines, summed from `ORDER_LINE_NET_SALES` in `EDLDB.BT_HCA_HCDM.ORDER_LINE_COST_MEASURES_RXP`, once per shipment-and-line record with a shipment or cancellation timestamp in the window. Split-shipment sales outside the window are excluded; records meeting both dates count once. Zero and negative sales remain included; any missing sales make the group total incomplete. Display rounds to two decimals; exports retain five. Current cancellation reasons are grouped case-insensitively; a missing reason does not establish an uncancelled order. Line-item status is `ORDER_ORDER_LINE_STATUS`; codes are shown as recorded. Conflicting statuses across shipment rows for a physical line fail validation. Orders can appear in multiple reason/status rows if their lines differ; TOTAL counts each order once. Line counts and sales are additive. Gray reason-row net-sales amounts have a recorded cancellation reason or status X and represent unrealized sales, not realized revenue. The gray TOTAL combines these amounts with the no-recorded-cancellation-reason, non-X amount. Recorded net sales are a baseline for opportunity sizing, not an estimate of recoverable revenue from cancelled orders.

3.1.2 What percentage of customers come back and place a successful vet diet order for the same pet and part number after an aged order in the next 70 days?

The three cohorts below compare non-cancelled reorders after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC. The yellow columns show returning customers. Each cohort uses its own fixed denominator.

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3.1.2.1 How do same-pet, same-product return rates and net sales compare across the three customer cohorts over 70 days?

Same-pet, same-product follow-up: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

Table 2: Same-pet, same-product follow-up: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

	All aged customers $N = 30,382$				First CUSTOMER_UPLOAD vet-diet order aged; $N = 11,550$				First vet-diet order aged $N = 5,097$			
Days	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales
7	2,316	7.62%	2,407	221,122.35	833	7.21%	855	72,184.56	312	6.12%	322	23,939.00
14	3,832	12.61%	4,143	386,124.68	1,281	11.09%	1,365	116,938.34	468	9.18%	500	36,606.75
21	5,133	16.89%	5,951	558,774.12	1,633	14.14%	1,840	159,141.20	565	11.08%	642	47,225.57
28	5,850	19.25%	7,289	685,508.89	1,846	15.98%	2,256	193,780.75	631	12.38%	792	58,032.36
35	6,461	21.27%	8,905	841,087.82	2,026	17.54%	2,698	234,112.61	680	13.34%	914	67,607.06
42	6,818	22.44%	10,118	958,460.32	2,149	18.61%	3,113	272,143.01	720	14.13%	1,061	79,250.13
49	7,280	23.96%	11,898	1,130,974.50	2,297	19.89%	3,641	322,633.43	749	14.69%	1,191	89,793.43
56	7,510	24.72%	13,152	1,252,292.42	2,372	20.54%	3,995	354,230.43	766	15.03%	1,308	99,811.18
63	7,756	25.53%	14,621	1,393,016.00	2,450	21.21%	4,425	393,797.77	781	15.32%	1,425	109,761.26
70	7,944	26.15%	15,808	1,506,797.98	2,509	21.72%	4,801	427,522.19	800	15.70%	1,567	120,434.49

Cohorts and anchors: All aged customers includes every qualifying aged cancellation, without a prescription-creation or first-order restriction. For each first-order cohort, the customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. The first customer-upload cohort uses the earliest observed vet-diet CUSTOMER_UPLOAD order across historical prescription sources, using linked prescriptions and ECOM order history. The first vet-diet cohort uses earliest observed vet-diet orders across all historical approval channels and prescription sources, using RxP vet-diet lines and ECOM order history. Each percentage divides returning customers by its group's fixed full cohort, including customers without a qualifying reorder or with unusable matching keys. The cohorts overlap; do not add their counts.
Shared follow-up rules: Require the same numeric customer ID, exact PRODUCT_PART_NUMBER and non-null PET_ID on a different order placed strictly after the matched cancellation through the indicated day, inclusive. Later orders may use any approval channel or prescription source within the certified RxP line view. Exclude the entire later order if any line has status X, a recorded cancellation reason, a cancellation timestamp or unknown line status. Statuses and prescription channels reflect the extraction snapshot. All windows require complete follow-up through 2026-10-28 00:00 UTC. Each customer, order and matching physical line counts once per window, despite repeated anchor associations. Windows overlap and must not be added. Net sales sum ORDER_LINE_NET_SALES once per matching shipment/line contribution, preserving zero and negative amounts; missing sales make totals incomplete. Sales can be recorded after the placement window and do not prove recovered revenue. Amounts display two decimals; exports retain five. N/A denotes a zero cohort denominator.

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3.1.3 What percentage of customers come back and place a successful order for the same pet but different product after an aged order in the next 70 days?

The three cohorts below compare non-cancelled reorders after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC. The yellow columns show returning customers. Each cohort uses its own fixed denominator.

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3.1.3.1 How do same-pet, different-product return rates and net sales compare across the three customer cohorts over 70 days?

Same-pet, different-product vet-diet follow-up: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

Table 3: Same-pet, different-product vet-diet follow-up: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

	All aged customers $N = 30,382$				First CUSTOMER_UPLOAD vet-diet order aged; $N = 11,550$				First vet-diet order aged $N = 5,097$			
Days	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales
7	1,332	4.38%	1,457	131,940.62	444	3.84%	466	40,916.08	181	3.55%	189	15,801.67
14	2,182	7.18%	2,553	230,737.09	727	6.29%	792	68,360.26	276	5.41%	307	24,589.86
21	2,890	9.51%	3,677	331,424.29	924	8.00%	1,089	94,359.87	332	6.51%	399	32,725.31
28	3,413	11.23%	4,690	423,723.90	1,080	9.35%	1,379	119,958.61	382	7.49%	505	40,896.95
35	3,814	12.55%	5,711	517,237.31	1,217	10.54%	1,695	147,193.63	423	8.30%	605	48,954.22
42	4,138	13.62%	6,656	602,642.69	1,314	11.38%	1,983	173,509.35	451	8.85%	704	57,525.03
49	4,421	14.55%	7,662	697,800.49	1,401	12.13%	2,289	201,286.86	479	9.40%	800	65,677.94
56	4,651	15.31%	8,624	784,480.62	1,479	12.81%	2,574	225,931.09	504	9.89%	896	73,577.63
63	4,856	15.98%	9,594	871,498.79	1,553	13.45%	2,893	253,413.39	530	10.40%	1,010	83,296.37
70	5,010	16.49%	10,552	958,728.45	1,617	14.00%	3,220	281,008.90	548	10.75%	1,122	91,928.98

Cohorts and anchors: All aged customers includes every qualifying aged cancellation, without a prescription-creation or first-order restriction. For each first-order cohort, the customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. The first customer-upload cohort uses the earliest observed vet-diet CUSTOMER_UPLOAD order across historical prescription sources, using linked prescriptions and ECOM order history. The first vet-diet cohort uses earliest observed vet-diet orders across all historical approval channels and prescription sources, using RxP vet-diet lines and ECOM order history. Each percentage divides returning customers by its group's fixed full cohort, including customers without a qualifying reorder or with unusable matching keys. The cohorts overlap; do not add their counts.
Shared follow-up rules: Require the same numeric customer ID and non-null PET_ID, a different nonblank PRODUCT_PART_NUMBER from the anchor line, and a TRUE vet-diet flag on the later line. Use a different order placed strictly after the matched cancellation through the indicated day, inclusive. Later orders may use any approval channel or prescription source within the certified RxP line view. Exclude the entire later order if any line has status X, a recorded cancellation reason, a cancellation timestamp or unknown line status. Statuses and prescription channels reflect the extraction snapshot. All windows require complete follow-up through 2026-10-28 00:00 UTC. Each customer, order and matching physical line counts once per window, despite repeated anchor associations. Windows overlap and must not be added. Net sales sum ORDER_LINE_NET_SALES once per matching shipment/line contribution, preserving zero and negative amounts; missing sales make totals incomplete. Sales can be recorded after the placement window and do not prove recovered revenue. Amounts display two decimals; exports retain five. N/A denotes a zero cohort denominator.

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3.1.4 What percentage of customers place a successful vet diet order after an aged order in the next 70 days?

The three cohorts below compare non-cancelled reorders after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC. The yellow columns show returning customers. Each cohort uses its own fixed denominator.

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3.1.4.1 How do successful vet-diet return rates and net sales compare across the three customer cohorts over 70 days?

Any vet-diet follow-up, for any pet or product: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

Table 4: Any vet-diet follow-up, for any pet or product: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

	All aged customers $N = 30,382$				First `CUSTOMER_UPLOAD` vet-diet order aged; $N = 11,550$				First vet-diet order aged $N = 5,097$			
Days	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales
7	3,594	11.83%	4,020	376,440.20	1,217	10.54%	1,305	115,132.09	438	8.59%	463	35,772.45
14	5,824	19.17%	7,052	666,309.41	1,914	16.57%	2,178	192,777.87	659	12.93%	737	55,857.40
21	7,637	25.14%	10,163	960,107.59	2,432	21.06%	3,005	267,367.22	800	15.70%	969	74,330.46
28	8,713	28.68%	12,680	1,202,534.61	2,766	23.95%	3,743	332,421.02	899	17.64%	1,208	92,469.85
35	9,555	31.45%	15,448	1,469,868.48	3,051	26.42%	4,540	405,245.66	980	19.23%	1,425	109,763.84
42	10,075	33.16%	17,772	1,693,731.72	3,240	28.05%	5,282	474,797.91	1,031	20.23%	1,653	128,552.70
49	10,631	34.99%	20,632	1,969,419.82	3,423	29.64%	6,127	555,407.00	1,072	21.03%	1,873	146,660.13
56	10,976	36.13%	23,017	2,198,126.45	3,547	30.71%	6,825	618,318.62	1,105	21.68%	2,077	163,846.89
63	11,314	37.24%	25,580	2,445,035.54	3,664	31.72%	7,624	693,046.86	1,137	22.31%	2,302	183,559.64
70	11,567	38.07%	27,878	2,665,017.53	3,753	32.49%	8,395	762,387.61	1,165	22.86%	2,541	202,269.09

Cohorts and anchors: All aged customers includes every qualifying aged cancellation, without a prescription-creation or first-order restriction. For each first-order cohort, the customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. The first customer-upload cohort uses the earliest observed vet-diet `CUSTOMER_UPLOAD` order across historical prescription sources, using linked prescriptions and ECOM order history. The first vet-diet cohort uses earliest observed vet-diet orders across all historical approval channels and prescription sources, using RxP vet-diet lines and ECOM order history. Each percentage divides returning customers by its group's fixed full cohort, including customers without a qualifying reorder or with unusable matching keys. The cohorts overlap; do not add their counts.
Shared follow-up rules: Require the same numeric customer ID and a TRUE vet-diet flag on the later line; purchases may be for any pet and any product part number. Use a different order placed strictly after the matched cancellation through the indicated day, inclusive. Later orders may use any approval channel or prescription source within the certified RxP line view. Exclude the entire later order if any line has status X, a recorded cancellation reason, a cancellation timestamp or unknown line status. Statuses and prescription channels reflect the extraction snapshot. All windows require complete follow-up through 2026-10-28 00:00 UTC. Each customer, order and matching physical line counts once per window, despite repeated anchor associations. Windows overlap and must not be added. Net sales sum `ORDER_LINE_NET_SALES` once per matching shipment/line contribution, preserving zero and negative amounts; missing sales make totals incomplete. Sales can be recorded after the placement window and do not prove recovered revenue. Amounts display two decimals; exports retain five. N/A denotes a zero cohort denominator.

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3.1.5 What percentage of customers place a Chewy order successfully in the following 70 days after their aged order?

The three cohorts below compare non-cancelled reorders after CMPU vet-diet `CUSTOMER_UPLOAD` aged cancellations from May 1, 2026 through July 31, 2026, inclusive, UTC. The yellow columns show returning customers. Each cohort uses its own fixed denominator.

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3.1.5.1 How do successful Chewy return rates and net sales compare across the three customer cohorts over 70 days?

All-Chewy follow-up, across all product categories: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

Table 5: All-Chewy follow-up, across all product categories: cumulative customers, cohort percentages, distinct orders and matching-line net sales across three aged-customer cohorts.

	All aged customers $N = 30,382$				First CUSTOMER_UPLOAD vet-diet order aged; $N = 11,550$				First vet-diet order aged $N = 5,097$			
Days	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales	Customers	Cohort (%)	Orders	Net sales
7	6,477	21.32%	8,512	702,653.54	2,198	19.03%	2,653	213,634.05	826	16.21%	971	74,076.57
14	10,265	33.79%	15,918	1,307,072.48	3,492	30.23%	4,913	391,195.10	1,310	25.70%	1,784	130,066.57
21	13,193	43.42%	23,645	1,912,382.98	4,589	39.73%	7,278	560,344.43	1,780	34.92%	2,685	179,905.10
28	14,804	48.73%	30,064	2,440,287.16	5,197	45.00%	9,308	718,308.85	2,035	39.93%	3,448	232,910.41
35	15,917	52.39%	36,535	2,970,675.24	5,610	48.57%	11,423	885,109.85	2,183	42.83%	4,142	282,786.69
42	16,689	54.93%	42,616	3,471,241.76	5,938	51.41%	13,515	1,054,750.51	2,306	45.24%	4,921	340,077.07
49	17,370	57.17%	49,096	4,009,256.32	6,201	53.69%	15,674	1,231,469.99	2,402	47.13%	5,643	393,935.24
56	17,835	58.70%	55,122	4,499,630.84	6,411	55.51%	17,688	1,392,251.83	2,482	48.70%	6,370	449,169.84
63	18,246	60.06%	61,142	4,993,682.85	6,574	56.92%	19,772	1,559,311.55	2,539	49.81%	7,064	501,643.27
70	18,564	61.10%	66,914	5,468,610.59	6,712	58.11%	21,905	1,730,599.16	2,591	50.83%	7,863	561,424.64

Cohorts and anchors: All aged customers includes every qualifying aged cancellation, without a prescription-creation or first-order restriction. For each first-order cohort, the customer's first qualifying aged order is the sole anchor; later aged orders do not reset follow-up. The first customer-upload cohort uses the earliest observed vet-diet CUSTOMER_UPLOAD order across historical prescription sources, using linked prescriptions and ECOM order history. The first vet-diet cohort uses earliest observed vet-diet orders across all historical approval channels and prescription sources, using RxP vet-diet lines and ECOM order history. Each percentage divides returning customers by its group's fixed full cohort, including customers without a qualifying reorder or with unusable matching keys. The cohorts overlap; do not add their counts.
Shared follow-up rules: Match the same numeric customer ID only; include all Chewy product categories and any pet or product. Use a different order placed strictly after the matched cancellation through the indicated day, inclusive. Follow-up orders come from EDLDB.ECOM.ORDER_LINE; no later prescription, approval-channel, prescription-source, vet-diet, product or pet restriction applies. Exclude the entire later order if any line or order has status X, a recorded cancellation reason, a cancellation timestamp or unknown line status. Statuses and prescription channels reflect the extraction snapshot. All windows require complete follow-up through 2026-10-28 00:00 UTC. Each customer, order and matching physical line counts once per window, despite repeated anchor associations. Windows overlap and must not be added. Net sales sum NET_SALES on all qualifying ECOM order lines once per physical order line, preserving zero and negative amounts. Exact source duplicates collapse and conflicting line values fail validation; missing sales make totals incomplete. Sales can be recorded after the placement window and do not prove recovered revenue. Amounts display two decimals; ECOM supplies four-decimal amounts, retained exactly in five-decimal exports. N/A denotes a zero cohort denominator.

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3.2 What is the contact pattern of CMPU aged order customers?

3.2.1 What is the distribution of CMPU customers with APPROVAL CHANNEL = CUSTOMER_UPLOAD that contacted Chewy by cancellation reason?

Table 6 compares customers with inbound contacts across CMPU vet-diet `CUSTOMER_UPLOAD` orders with qualifying shipment or cancellation activity from May 1, 2026 through July 31, 2026, inclusive, UTC.

3.2.1.1 What proportion of customers contacted Chewy for each order cancellation reason?

Distinct customers contacting by current order cancellation reason, through the latest observed shipment or cancellation event.

Table 6: Distinct customers contacting by current order cancellation reason, through the latest observed shipment or cancellation event.

Cancellation reason	Cohort customers	Customers contacting	Contacted (%)
[SQL NULL]	172,878	51,983	30.07%
Aged order	30,382	30,304	99.74%
Customer initiated cancellation	29,455	10,567	35.88%
Customer request	11,382	9,377	82.38%
Duplicate order	161	111	68.94%
Fraud	36	13	36.11%
Inventory issue	5	2	40.00%
No response	1,035	898	86.76%
Other	784	661	84.31%
Payment declined	4,242	2,393	56.41%
Payment issue	46	32	69.57%
Pharmacy rejection	278	264	94.96%
Prescription issue	5,051	5,012	99.23%
Restricted availability	1	1	100.00%
TOTAL	225,434	101,227	44.90%

Notes: No lower or upper day limit relative to order creation applies. Include contacts at any available earlier timestamp, capped inclusively at the latest observed shipment/cancellation timestamp among qualifying rows for that order and reason. For split or mixed outcomes, use the later of the observed events. Match inbound contacts on the original order ID and numeric customer ID, across all categories. Include handled, unhandled and unknown-handled inbound attempts. Count each customer once per reason across all their qualifying orders and contact days. Percentages divide contacting customers by all eligible customers in that reason, including those without a qualifying contact. Customers can appear in multiple reason rows; TOTAL independently counts each customer once overall and is not the sum of the reason rows. Reasons reflect the current snapshot, grouped case-insensitively after trimming. SQL NULL is explicit; blank strings, if present, form a separate row. NULL does not imply no cancellation. This cohort includes 434 NULL-reason order/reason records with observed cancellation but no qualifying shipment. This summary has no creation-relative day bounds. Observation ends at each order's observed event, so exposure varies; these are descriptive contact shares, not causal effects. Exact duplicate contacts collapse; missing or mismatching contact customers, missing contact times and non-inbound directions are excluded and audited. Inbound direction is a source proxy, not proof of human initiation. No prescription-creation or first-order restriction applies. Extraction time (UTC): 2026-11-04T13:31:51.835431+00:00. Executable SQL and complete order/contact records are under `data/opportunity/customer-contacts-by-cancel-reason/`.

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3.2.2 When did the CMPU aged order customers contact Chewy relative to the cancellation date?

Table 7 summarizes inbound contacts linked to CMPU vet-diet `CUSTOMER_UPLOAD` aged orders cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC. The full cohort contains 30,382 customers and 38,314 aged orders.

3.2.2.1 How many customers contacted Chewy on each day relative to aged-order cancellation?

Customer-initiated inbound contacts by day relative to aged-order cancellation: distinct customers contacting and distinct aged order IDs, from 13 days before through day 3 after cancellation. A total of 30,299 distinct customers contacted across this window.

Table 7: Customer-initiated inbound contacts by day relative to aged-order cancellation: distinct customers contacting and distinct aged order IDs, from 13 days before through day 3 after cancellation. A total of 30,299 distinct customers contacted across this window.

Days relative to cancellation	Customers contacting	Aged order IDs
-13	38	38
-12	858	861
-11	19,267	23,158
-10	11,364	13,530
-9	571	575
-8	215	215
-7	143	143
-6	123	123
-5	106	106
-4	99	99
-3	128	128
-2	251	256
-1	248	248
0	71	71
1	53	53
2	28	28
3	22	23

Notes: Customer-initiated is defined by `SESSION_INITIAL_DIRECTION=inbound` in `EDLDB.BT_CUSTOMER_CARE.FCT_CONTACTS_BRD`, across all contact categories. Include handled, unhandled and unknown-handled inbound attempts. Match the original aged `ORDER_ID` and require the contact's numeric `CUSTOMER_ID` to match the order's customer. Use UTC `SESSION_START_DTTM`. Day n is the elapsed 24-hour interval $[n, n+1)$ days relative to cancellation: day -1 is the preceding 24 hours, and day 0 starts exactly at cancellation. These are elapsed bins rather than calendar-date differences. All seventeen rows from -13 through $+3$ are retained, with zero for missing days. Count each customer and aged order once per day, regardless of repeat contact legs, sessions or qualifying order lines. Counts may overlap across days and should not be added as global unique totals. Missing timestamps or customer IDs, mismatching customer IDs, unknown directions and outbound contacts remain audited and are excluded from this table. Email contacts with Other ingress are included; inbound direction is a source proxy and does not establish human initiation. Full channel/ingress audits are exported with the data. Original eligibility uses cancellation dates and all qualifying orders, without a first-order or prescription-creation restriction. Observation cutoff: 2026-10-28 00:00 UTC, allowing the full positive-day window.

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3.2.3 What was the frequency of contacting Chewy for CMPU aged order customers?

Table 8 groups customers by the number of distinct relative days on which they contacted during the same -13 through $+3$ window.

3.2.3.1 How many distinct days did aged-order customers contact Chewy?

Frequency of customer contact days around aged-order cancellation: number of distinct relative days contacted versus number of customers. Of 30,382 cohort customers, 30,299 contacted on at least one day.

Table 8: Frequency of customer contact days around aged-order cancellation: number of distinct relative days contacted versus number of customers. Of 30,382 cohort customers, 30,299 contacted on at least one day.

Number of days contacted	Number of customers
0	83
1	27,265
2	2,808
3	201
4	24
5	1
6	0
7	0
8	0
9	0
10	0
11	0
12	0
13	0
14	0
15	0
16	0
17	0

Notes: Same CMPU vet-diet `CUSTOMER_UPLOAD` aged cohort, cancellation dates, observation cutoff and inbound contact rules as Table 7. Count each distinct relative 24-hour day bin from -13 through $+3$ once per customer across all qualifying aged orders. Multiple contacts, sessions, legs or orders on the same relative day do not increase the day count. These are relative cancellation-day bins, rather than distinct calendar dates. Each customer appears in exactly one frequency row; zero includes cohort customers with no qualifying contact in this window. All frequencies from 0 through 17 are retained, including zero-count rows. Customer counts sum to the full cohort; positive-day rows sum to the distinct contacting customers.

3.2.4 Does more frequent contact imply higher propensity to re-order successfully for the same pet and part number in 14 days?

Table 9 applies the contact-day frequency segments in Table 8 to same-pet, same-product reorders.

3.2.4.1 How do 14-day same-pet, same-product return rates vary with contact-day frequency?

Same-pet, same-product-part-number reorders within 14 days after aged cancellation, by customer contact-day frequency. Original CMPU vet-diet `CUSTOMER_UPLOAD` orders were cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC.

Table 9: Same-pet, same-product-part-number reorders within 14 days after aged cancellation, by customer contact-day frequency. Original CMPU vet-diet `CUSTOMER_UPLOAD` orders were cancelled from May 1, 2026 through July 31, 2026, inclusive, UTC.

Days contacted	Segment customers	Customers reordered	Segment reordered (%)
1	27,265	3,376	12.38%
2	2,808	396	14.10%

Notes: Segments count distinct relative 24-hour contact-day bins from -13 through $+3$, pooled across each customer's qualifying aged orders. Multiple contacts on the same relative day count as one day. Match inbound contacts to the original aged order and its customer, across all categories. Include handled, unhandled and unknown-handled inbound attempts. Reorders must match the customer, non-null `PET_ID` and nonblank `PRODUCT_PART_NUMBER`, on a different order placed strictly after its matched aged cancellation through 14 days inclusive. Apply the established non-cancelled-order rules: exclude an entire reorder order if any line has cancellation evidence or unknown status. Reorders may use any approval channel or prescription source. Count each customer once in their segment if any qualifying aged order has a matching reorder in its own follow-up window. Percentages use all customers in the contact segment, including those missing matching keys or without reorders; N/A denotes an empty segment. Contact-day segments include post-cancellation contacts through day $+3$. Observation cutoff: 2026-10-28 00:00 UTC.

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3.2.5 Does contact closer to cancellation imply higher propensity to re-order successfully in 14 days for the same pet and part number?

Table 10 compares customers with a qualifying contact on days -7 through $+3$ relative to aged cancellation against the rest of the aged-order cohort. The contacting group's 14-day reorder rate is 7.70 percentage points higher.

3.2.5.1 How do 14-day same-pet, same-product return rates compare for customers contacting on days -7 through +3 and other customers?

Same-pet, same-product-part-number reorders within 14 days after aged cancellation: customers contacting on days -7 through $+3$ versus the rest of the cohort. Original CMPU vet-diet `CUSTOMER_UPLOAD` orders were cancelled May 1–July 31, 2026, inclusive, UTC.

Table 10: Same-pet, same-product-part-number reorders within 14 days after aged cancellation: customers contacting on days -7 through $+3$ versus the rest of the cohort. Original CMPU vet-diet `CUSTOMER_UPLOAD` orders were cancelled May 1–July 31, 2026, inclusive, UTC.

Customer group	Customers	Customers reordered	Reorder rate (%)
Contacted on days -7 through $+3$	1,184	237	20.02%
Rest of aged cohort	29,198	3,595	12.31%

Notes: A customer belongs to the contacting group if any qualifying inbound contact is linked to one of their original aged orders and the contact customer ID matches the order's customer. Include all contact categories and handled states. Days are elapsed 24-hour bins: the window starts exactly seven days before cancellation and ends just before four days after cancellation, covering all of day $+3$. The complementary group includes customers contacting only outside this window and customers with no qualifying contacts. Each customer belongs to exactly one group; the groups sum to 30,382 customers and 3,832 returning customers. Use the same non-cancelled same-customer, `PET_ID` and `PRODUCT_PART_NUMBER` reorder endpoint as Table 9: a different order placed strictly after its matched aged cancellation through 14 days inclusive. Count a customer once if any qualifying aged order has a matching reorder in its own window; the contact need not concern that same aged order. Percentages use all customers in each group. Observation cutoff: October 28, 2026 at 00:00 UTC. Contacts after cancellation contribute to group membership; this is a descriptive comparison and does not establish a causal effect of contact.

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3.3 What are the delays in processing uploaded prescriptions?

An order becomes an aged order when canceled automatically after the time between order creation and time of shipment or any other cancellation exceeds 11 days. A detailed breakdown of cancellation reasons can be found in Figure 1 for the study population.

The population of orders for this study consists of first orders and first order lines for prescriptions as we would like to analyze issues at prescription upload and approval.

For this analysis; 95,106 vet diet prescriptions created between August 15, 2026 and October 15, 2026 were used with `CUSTOMER_UPLOAD` as the approval channel. This corresponds to 87,757 distinct orders and 95,106 order lines. The 24,408 orders described in `OMS` with the cancellation reason `AGED_ORDER` constitute the aged orders leading to 26,443 line items, 26,443 corresponding prescriptions and 20,587 customers.

The difference in counts between first orders and order lines and all orders and order lines is shown in Table 13.

Factors that might be contributing to an order aging are: delay in prescription upload by the customer, delay in processing the upload, and delay in approving the prescription after the uploaded prescription document is accepted.

Matched upload coverage is 29,655 of 95,106 study first-order lines (31.18%), compared with 777 of 26,443 aged first-order lines (2.94%); 25,666 aged lines (97.06%) have no matched upload in the retained window (Table 14). The matched lines link to 29,750 distinct upload entities overall and 782 in the aged subset (Table 11). These are different counting units: one entity can link to multiple prescriptions or lines. Unmatched evidence does not establish customer non-response, and uploads are not required to precede cancellation.

Of the 782 matched aged-order upload entities, 52 (6.65%) are currently ACCEPTED. Separately, 6 of 26,443 aged prescriptions (0.02%) have a recorded approval (Table 12). These snapshot counts do not form a same-grain conversion funnel from upload through approval. The median time for the accepted entities to become accepted was 10 days after upload, as a result missing the aged order cancellation deadline (Figure 4). For comparison, the distributions of the time difference between upload of the documents and their final states are in Figure 3 for all entities. Customers seem to upload at similar delays after order creation for overall and aged order prescriptions (Figure 2).

From the available entity upload data, it takes less than a minute to approve an accepted prescription document see Figure 5.

3.3.1 Which prescriptions, orders, customers, and uploads are included in the study?

3.3.1.1 How many prescriptions, orders, lines, customers, and uploaded documents are in each study population?

Study population and uploaded-document counts. Counts are distinct within each population.

Table 11: Study population and uploaded-document counts. Counts are distinct within each population.

					Uploaded documents	
Population	Prescriptions	Orders	Order lines	Customers	Total	Accepted
All approval channels, matched to first-order lines	103,550	95,567	103,550	80,651	30,770	26,294
`CUSTOMER_UPLOAD`, matched to first-order lines	95,106	87,757	95,106	73,906	29,750	25,645
`CUSTOMER_UPLOAD` aged-cancellation subset	26,443	24,408	26,443	20,587	782	52

Notes: CMPU vet-diet prescriptions created from August 15, 2026 inclusive through October 15, 2026 exclusive, UTC. The aged-cancellation subset includes qualifying cancellations observed through October 27, 2026, 00:00 UTC. Line associations and document counts were refreshed on November 04, 2026. Uploaded documents count distinct linked `DIET_PRESCRIPTION` upload entities created August 1–October 25, 2026, UTC, with a matched first-order line; entities shared by prescriptions count once per row. Accepted documents have current `STATE_NAME = ACCEPTED`.

3.3.1.2 How are prescription approval decisions distributed across the study populations?

Prescription decisions for the study populations.

Table 12: Prescription decisions for the study populations.

Population	Prescriptions	Approved	Unknown	Rejected
All approval channels, matched to first-order lines	103,550	36,908	66,595	47
`CUSTOMER_UPLOAD`, matched to first-order lines	95,106	30,537	64,569	0
`CUSTOMER_UPLOAD` aged-cancellation subset	26,443	6	26,437	0

Notes: Distinct prescriptions in the same populations as Table 11. Approved means `PRESCRIPTION_APPROVED = TRUE`, consistent with the prescription registry's `APPROVED_NUMERATOR`. Rejected means current `RX_STATUS = DENIED` without an approval flag. Unknown includes cancelled and pending prescriptions without a recorded approval or rejection outcome. The three categories are mutually exclusive and sum to the prescription count in each row. Decisions use the current source snapshot refreshed November 04, 2026; approval and denial times are not restricted to the aged-cancellation cutoff.

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3.3.1.3 How do cancellation reasons, order counts, and net sales compare between first orders and all linked orders?

Cancellation reasons, order counts and net sales for first orders and all linked orders.

Table 13: Cancellation reasons, order counts and net sales for first orders and all linked orders.

	First orders			All orders		
Cancellation reason	Orders	Lines	Net sales	Orders	Lines	Net sales
[Aged order outside cutoff]	372	398	33,489.70	372	398	33,489.70
[No recorded cancellation reason]	38,424	40,870	3,161,976.99	64,865	73,958	5,748,033.82
Aged order	24,408	26,443	2,236,727.20	24,409	26,445	2,236,909.55
Customer initiated cancellation	11,603	13,129	995,465.64	12,381	14,083	1,076,754.83
Customer request	4,891	5,487	428,945.72	5,049	5,698	448,008.69
Duplicate order	61	64	5,237.98	76	82	6,635.26
Fraud	13	15	1,809.92	13	15	1,809.92
Inventory issue	3	3	264.42	3	3	264.42
No response	541	575	48,509.38	565	600	50,813.21
Other	302	336	25,662.96	318	355	26,944.01
Payment declined	2,449	2,688	221,833.07	2,567	2,821	232,508.70
Payment issue	15	17	1,503.07	17	19	1,694.00
Pharmacy rejection	150	171	14,278.29	150	171	14,278.29
Prescription issue	4,524	4,909	412,858.46	4,524	4,910	412,972.44
Reseller	1	1	132.99	1	1	132.99
TOTAL	87,757	95,106	7,588,695.79	115,310	129,559	10,291,249.83

Notes: CMPU vet-diet `CUSTOMER_UPLOAD` prescriptions created August 15–October 14, 2026, UTC. All orders includes first and subsequent linked orders, without an order-placement window. Qualifying aged cancellations are observed on or before October 27, 2026, 00:00 UTC; current aged reasons outside that cutoff are shown separately for reconciliation. Other reasons reflect the comparison snapshot. Counts are distinct orders and physical lines within each group. Net sales sum `ORDER_LINE_NET_SALES` once per shipment/line record, for qualifying lines only. Amounts are displayed to two decimals; five-decimal values remain in the data exports. Gray net-sales amounts have a recorded cancellation reason and represent unrealized sales, not realized revenue. TOTAL includes these gray amounts. The financial cohort is derived from current prescription/order records and reconciled to the current first-order population.

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3.3.1.4 How do first-order line statuses relate to accepted prescription uploads?

CMPU vet-diet first-order lines: `CUSTOMER_UPLOAD` approval channel.

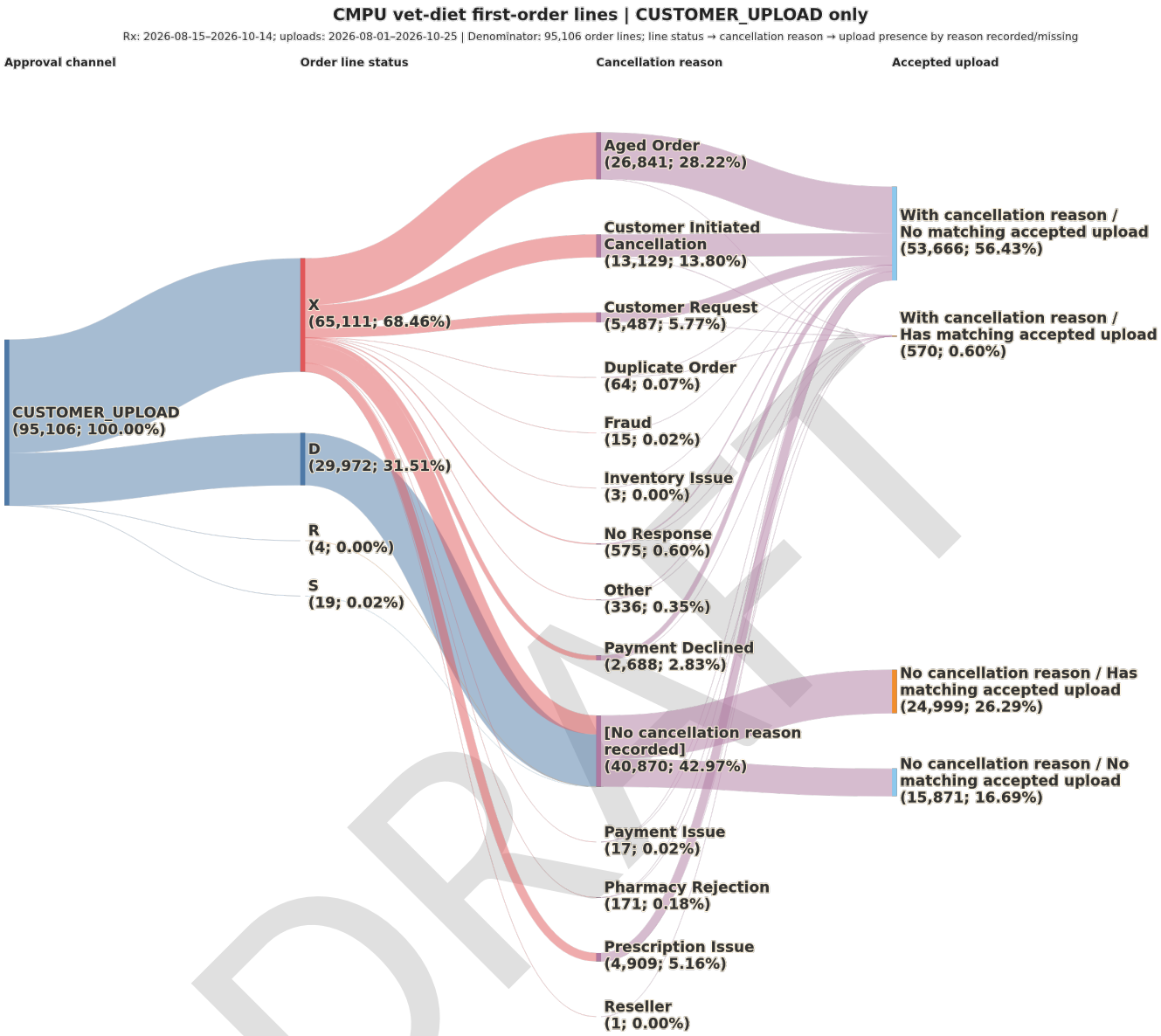


Figure 1: CMPU vet-diet first-order lines: `CUSTOMER_UPLOAD` approval channel.

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3.3.2 What is the percentage of prescription upload?

Table 14 reports upload coverage for the study's first orders and first-order lines, by cancellation reason and line-item status.

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3.3.2.1 What proportion of first-order lines have prescription uploads by cancellation reason and line-item status?

Prescription upload coverage by first-order cancellation reason and line-item status: CMPU vet-diet CUSTOMER_UPLOAD prescriptions created August 15–October 14, 2026, inclusive, UTC.

Table 14: Prescription upload coverage by first-order cancellation reason and line-item status: CMPU vet-diet CUSTOMER_UPLOAD prescriptions created August 15–October 14, 2026, inclusive, UTC.

Cancellation reason	Line-item status	First orders	Prescriptions	First-order line items	Line items with upload	Upload coverage (%)	Total uploads	Accepted uploads	Rejected uploads
[Aged order outside cutoff]	X	372	398	398	14	3.52%	14	0	9
[No recorded cancellation reason]	D	28,996	29,972	29,972	24,991	83.38%	25,069	24,962	94
[No recorded cancellation reason]	R	4	4	4	4	100.00%	4	4	0
[No recorded cancellation reason]	S	16	19	19	17	89.47%	17	17	0
[No recorded cancellation reason]	X	9,780	10,875	10,875	1,290	11.86%	1,299	92	251
Aged order	X	24,408	26,443	26,443	777	2.94%	782	52	194
Customer initiated cancellation	X	11,603	13,129	13,129	1,160	8.84%	1,161	311	150
Customer request	X	4,891	5,487	5,487	1,082	19.72%	1,083	145	174
Duplicate order	X	61	64	64	14	21.88%	14	6	2
Fraud	X	13	15	15	0	0.00%	0	0	0
Inventory issue	X	3	3	3	0	0.00%	0	0	0
No response	X	541	575	575	9	1.57%	9	8	0
Other	X	302	336	336	74	22.02%	74	16	8
Payment declined	X	2,449	2,688	2,688	15	0.56%	15	13	1
Payment issue	X	15	17	17	1	5.88%	1	1	0
Pharmacy rejection	X	150	171	171	14	8.19%	14	3	1
Prescription issue	X	4,524	4,909	4,909	193	3.93%	194	15	47
Reseller	X	1	1	1	0	0.00%	0	0	0
TOTAL	All	87,757	95,106	95,106	29,655	31.18%	29,750	25,645	931

Notes: Coverage = 100 × distinct first-order lines with at least one matched upload ÷ all study first-order lines in that reason/status group, including lines without uploads. Prescriptions count distinct `RX_IDs` and equal line-item counts in this validated one-prescription-per-first-order-line cohort. Orders and physical order/line pairs count distinctly; shipment rows cannot multiply counts. The validated population is the same as Table 11. Uploads are distinct `UPLOAD_IDs` from HCDM `rxp_upload_entity`, restricted to `DIET_PRESCRIPTION` entities. Expand prescription IDs and match both prescription and numeric customer IDs, then both first-order keys. Upload discovery is August 1–October 25 inclusive, UTC; missing creation timestamps follow the existing study's unknown-window rule and are audited. No requirement that upload precede cancellation is imposed. Accepted and rejected refer to current entity `STATE_NAME=ACCEPTED` and `REJECTED`, not prescription approval or denial. Other entity states remain in Total uploads, so Accepted plus Rejected need not equal Total. Reasons are trimmed and case-insensitive. Aged order uses the inclusive October 27, 2026, 00:00 UTC cancellation cutoff; later or undated aged reasons remain in the outside-cutoff row. Missing/blank reasons do not prove an uncancelled order. Line-item status is `ORDER_ORDER_LINE_STATUS` from the coordinated first-order-line extract, with shipment-row consistency checked. Codes are shown as recorded; missing status is explicit. Table 15 uses reconciled status/reason evidence. An entity may match multiple prescriptions/lines and reason/status groups; an order can also span groups. TOTAL independently deduplicates both and is not the sum of overlapping rows. Cancellation reasons/times use the retained prescription-decision extract and agree with the upload and status extracts. Counts describe the validated source snapshots.

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3.3.2.2 How do prescription, order, and line-item statuses vary by upload coverage when no cancellation reason is recorded?

Prescription and order statuses for first-order lines with no currently recorded cancellation reason, by retained upload coverage.

Table 15: Prescription and order statuses for first-order lines with no currently recorded cancellation reason, by retained upload coverage.

RX_STATUS	Order status	Order cancellation reason	Line-item status	Prescriptions	Line items	With upload	Without upload
APPROVED	C	[No recorded cancellation reason]	R	1	1	1	0
APPROVED	C	[No recorded cancellation reason]	S	17	17	15	2
APPROVED	D	[No recorded cancellation reason]	D	28,498	28,498	23,928	4,570
APPROVED	D	[No recorded cancellation reason]	X	21	21	18	3
APPROVED	R	[No recorded cancellation reason]	R	3	3	3	0
APPROVED	S	[No recorded cancellation reason]	S	1	1	1	0
APPROVED	X	[No recorded cancellation reason]	X	9	9	8	1
CLOSED	C	[No recorded cancellation reason]	S	1	1	1	0
CLOSED	D	[No recorded cancellation reason]	D	1,442	1,442	1,058	384
CLOSED	D	[No recorded cancellation reason]	X	24	24	22	2
CLOSED	X	[No recorded cancellation reason]	X	1	1	1	0
SYSTEM_CLOSED	C	[No recorded cancellation reason]	X	70	70	1	69
SYSTEM_CLOSED	D	[No recorded cancellation reason]	D	32	32	5	27
SYSTEM_CLOSED	D	[No recorded cancellation reason]	X	10,553	10,553	1,220	9,333
SYSTEM_CLOSED	R	[No recorded cancellation reason]	X	2	2	0	2
SYSTEM_CLOSED	S	[No recorded cancellation reason]	X	1	1	0	1
SYSTEM_CLOSED	X	[No recorded cancellation reason]	X	192	192	20	172
VET_REVIEW	D	[No recorded cancellation reason]	X	2	2	0	2
Total				40,870	40,870	26,302	14,568

Notes: Start with the validated 40,870 prescriptions/first-order lines (38,424 orders) in the no-recorded-reason rows of Table 14. At the status extraction (2026-11-04T13:34:44.605000+00:00), 40,870 lines still had a null or blank reason; 0 lines with newly populated reasons are excluded and retained in the audit. Prescriptions count distinct `RX_ID`s and equal line-item counts in this one-prescription-per-first-order-line cohort. Each physical first-order line counts once. Validate both first-order keys and customer identity; collapse shipment rows only when their customer, statuses and reason agree. Status codes are shown as recorded; a missing cancellation reason does not establish that an order shipped successfully.

With/without upload counts refer to lines with/without at least one matched entity, not the number of entities. Reuse the validated HCDM diet-prescription upload associations from the preceding table: match prescription ID, customer ID, order ID and line ID. The August 1–October 25, 2026 inclusive UTC discovery window, unknown-creation-time rule and all entity states are retained. Uploads need not precede cancellation. Uploads and status evidence are reconciled.

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3.3.2.3 How do prescription, order, and line-item statuses vary by upload coverage for aged orders?

Prescription and order statuses for aged-order first-order lines, by retained upload coverage.

Table 16: Prescription and order statuses for aged-order first-order lines, by retained upload coverage.

RX_STATUS	Order status	Order cancellation reason	Line-item status	Prescriptions	Line items	With upload	Without upload
APPROVED	X	aged order	X	5	5	4	1
CLOSED	X	aged order	X	1	1	1	0
SYSTEM_CLOSED	X	aged order	X	26,436	26,436	772	25,664
VET_REVIEW	X	aged order	X	1	1	0	1
Total				26,443	26,443	777	25,666

Notes: Same columns and counting rules as Table 15, for the frozen aged-order subset of Table 14: 26,443 prescriptions/first-order lines across 24,408 orders. Include cancellations at or before October 27, 2026, 00:00 UTC; later or undated aged cancellations are excluded. Statuses are from the retained extraction at 2026-11-04T13:34:44.605000+00:00. Prescriptions count distinct `RX_IDs`; each physical line counts once. With/without upload counts lines with/without at least one entity, not entities. Validate prescription/customer/both first-order keys and shipment-row consistency. Reuse all entity states and the August 1–October 25 inclusive UTC discovery window, including the retained unknown-creation-time rule. Uploads need not precede cancellation.

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3.3.3 How long do customers wait to upload a prescription?

Figure 2 compares upload entity creation relative to order placement for the overall and aged-order populations.

The distribution of time between document upload for the whole analysis population and the aged orders do not differ significantly.

3.3.3.1 How does time from order creation to upload compare between all study uploads and aged-order uploads?

Upload creation relative to order placement for the overall and aged-order entity populations. Elapsed time is upload creation minus order placement: positive values indicate upload after ordering and negative values indicate upload before ordering. Each entity uses its earliest matched first-order placement. Plotted durations include 29,715 overall entities and 782 aged-order entities; entities missing an order-placement timestamp are excluded from the boxplots. Medians, quartiles, 1.5-times-IQR whiskers and all outliers are shown.

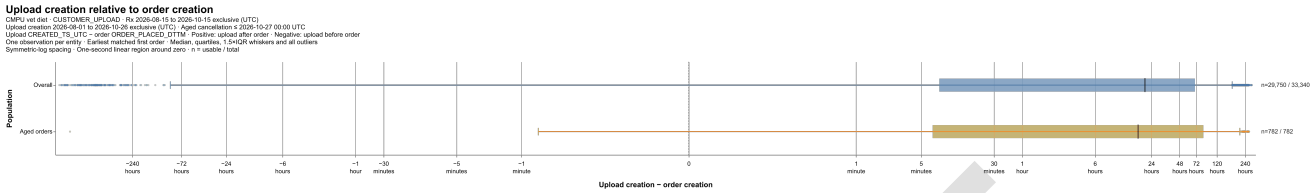


Figure 2: Upload creation relative to order placement for the overall and aged-order entity populations. Elapsed time is upload creation minus order placement: positive values indicate upload after ordering and negative values indicate upload before ordering. Each entity uses its earliest matched first-order placement. Plotted durations include 29,715 overall entities and 782 aged-order entities; entities missing an order-placement timestamp are excluded from the boxplots. Medians, quartiles, 1.5-times-IQR whiskers and all outliers are shown.

3.3.4 How long does it take for an uploaded prescription to be ready for approval?

Figures 3 and 4 show the elapsed time from upload entity creation to its current recorded state change, grouped by final state, for the overall study-linked population and its aged-order subset. Both figures use the same state order and time scale.

3.3.4.1 How long does it take study uploads to reach their recorded states?

Upload creation to current recorded state change for all study-linked CMPU vet-diet `CUSTOMER_UPLOAD` entities ($N = 33{,}336$). Each upload entity counts once. Boxplots show medians, quartiles, 1.5-times-IQR whiskers and all outliers, grouped by final state.

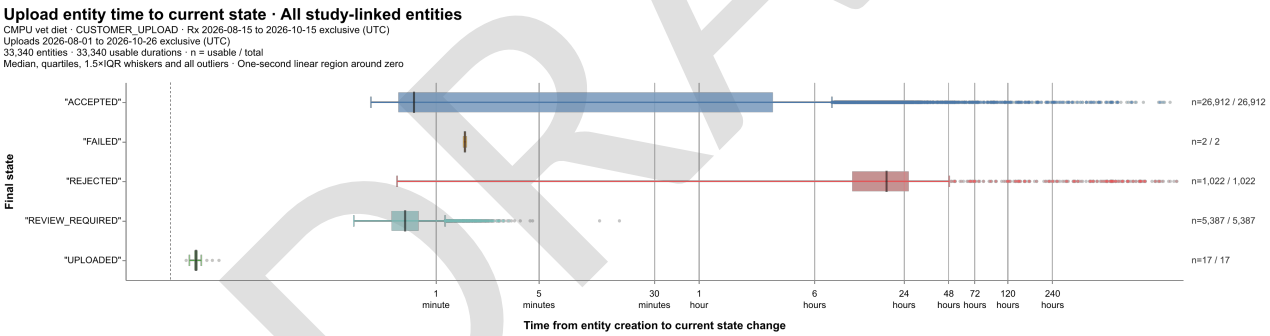
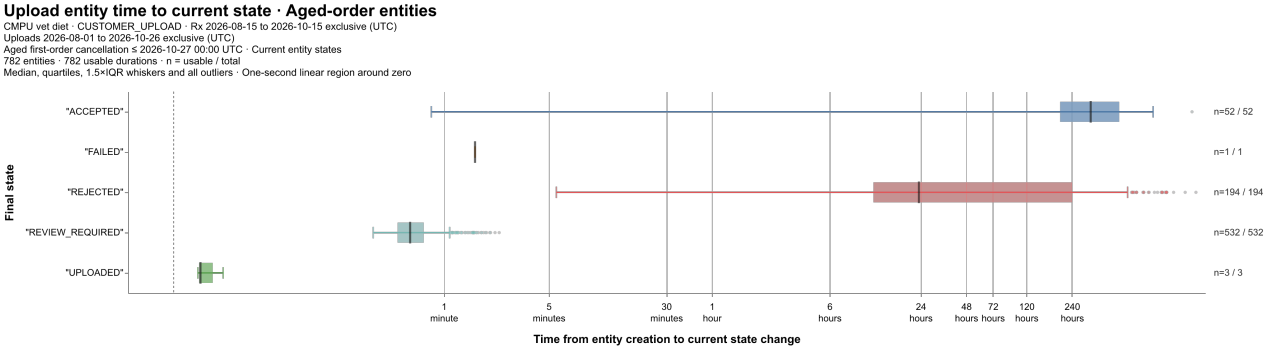


Figure 3: Upload creation to current recorded state change for all study-linked CMPU vet-diet `CUSTOMER_UPLOAD` entities ($N = 33{,}336$). Each upload entity counts once. Boxplots show medians, quartiles, 1.5-times-IQR whiskers and all outliers, grouped by final state.

3.3.4.2 How long does it take aged-order uploads to reach their recorded states?

Upload creation to current recorded state change for the aged-order entity subset ($N = 782$). A qualifying matched first-order line was cancelled for aging on or before October 27, 2026, 00:00 UTC. Current entity states and state-change times are not capped at cancellation.



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Figure 4: Upload creation to current recorded state change for the aged-order entity subset ($N = 782$). A qualifying matched first-order line was cancelled for aging on or before October 27, 2026, 00:00 UTC. Current entity states and state-change times are not capped at cancellation.

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3.3.5 How long does it take for an accepted prescription document to get approved?

Figure 5 shows the cumulative distribution of the interval between the selected upload entity's accepted-state change and the latest recorded approval for qualifying first-order lines.

3.3.5.1 How long after an accepted upload is the prescription approved?

Cumulative distribution of accepted-upload-to-approval time for CMPU vet-diet `CUSTOMER_UPLOAD` first-order lines with current status D. Elapsed time is the latest recorded approval timestamp minus the selected accepted entity's state-change timestamp. The curve shows only nonnegative gaps from 0 through 60 seconds inclusive and is normalized to that subset; gaps above 60 seconds are omitted. Each distinct matched first-order line counts once.

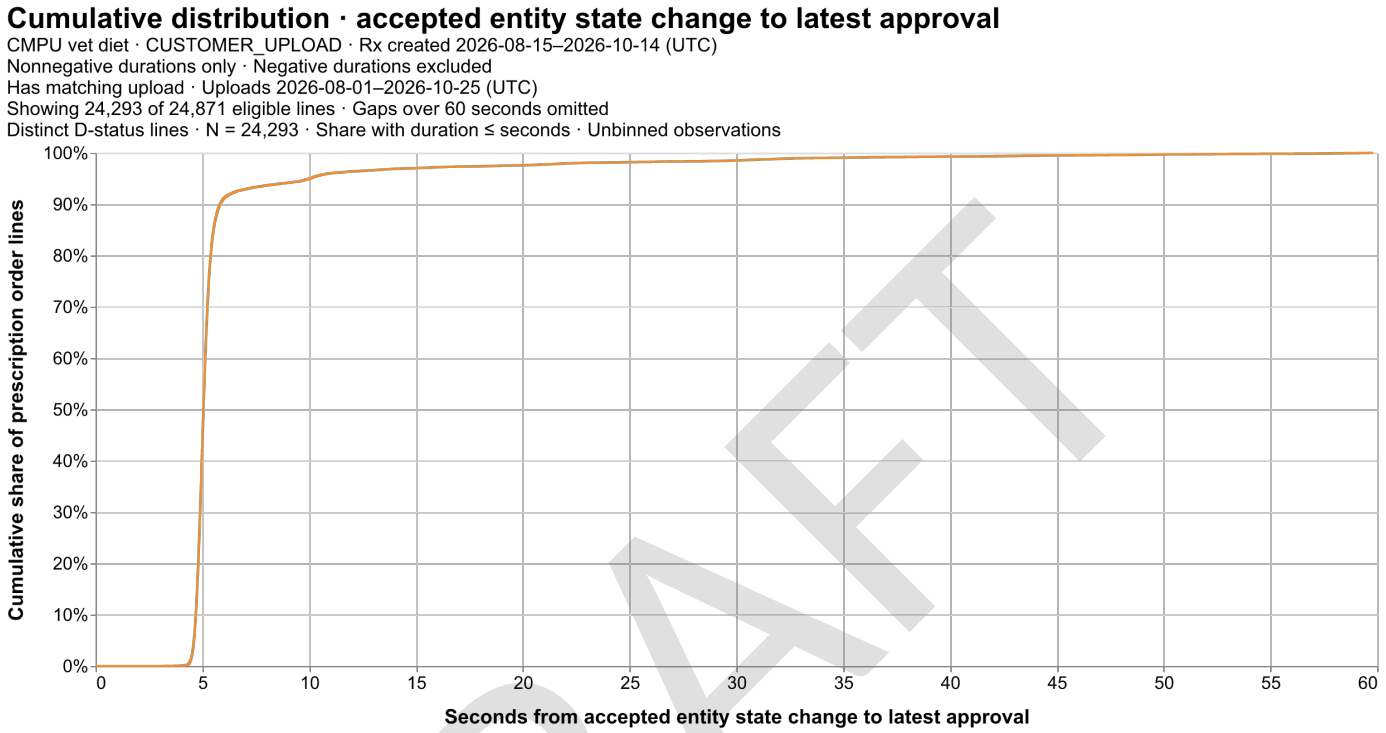


Figure 5: Cumulative distribution of accepted-upload-to-approval time for CMPU vet-diet `CUSTOMER_UPLOAD` first-order lines with current status D. Elapsed time is the latest recorded approval timestamp minus the selected accepted entity's state-change timestamp. The curve shows only nonnegative gaps from 0 through 60 seconds inclusive and is normalized to that subset; gaps above 60 seconds are omitted. Each distinct matched first-order line counts once.

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3.4 What is the probability that after X days the order will be an aged order?

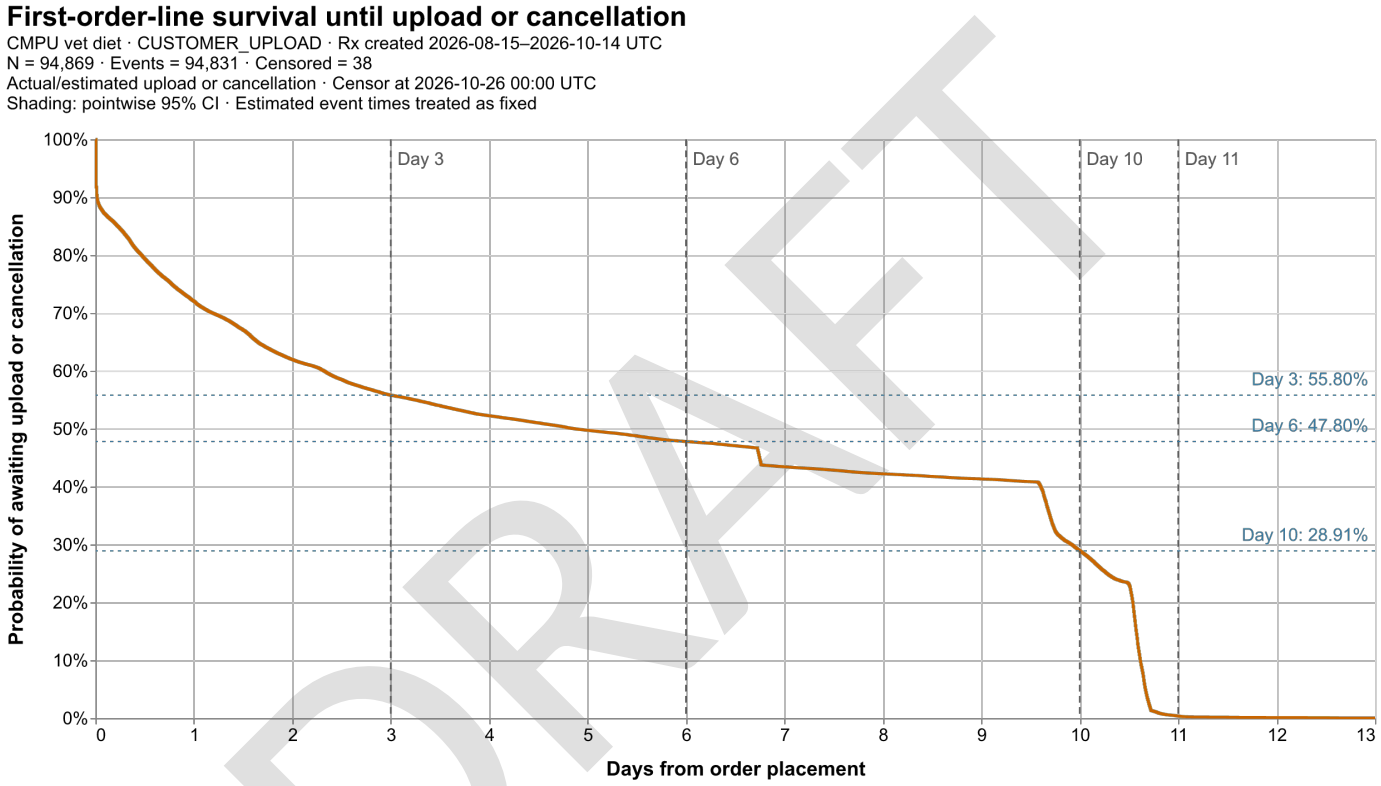
We look at the chance of an order becoming an aged order in this section. Here, we build a survival curve. Survival means that an order has yet to encounter a accepted document upload or a cancellation. The The cancellations after day 10 are system created aged order cancellations. From figure 6, that after day 3, the change of an order becoming an aged order is about 50% (28.90%/55.80%) if there is no accepted document upload or any cancellation except aged order.

Figure 6 shows the probability that a first-order line is still awaiting an accepted upload or cancellation after order placement.

3.4.1 How long do first-order lines remain without an accepted upload or cancellation?

3.4.1.1 What proportion of first-order lines remain without an accepted upload or cancellation over time?

Kaplan–Meier survival curve for 94,869 eligible distinct first-order lines linked to CMPU vet-diet `CUSTOMER_UPLOAD` prescriptions created August 15–October 14, 2026, inclusive, UTC. Time zero is order placement; the event is the earlier of the actual or estimated accepted-entity state-change time and order cancellation. The upload-entity discovery window is August 1–October 25, 2026, inclusive. Missing events and events after October 26, 2026 at 00:00 UTC are right-censored at that cutoff; events exactly at the cutoff are observed. Survival means neither accepted-state change nor cancellation has occurred.



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Figure 6: Kaplan–Meier survival curve for 94,869 eligible distinct first-order lines linked to CMPU vet-diet `CUSTOMER_UPLOAD` prescriptions created August 15–October 14, 2026, inclusive, UTC. Time zero is order placement; the event is the earlier of the actual or estimated accepted-entity state-change time and order cancellation. The upload-entity discovery window is August 1–October 25, 2026, inclusive. Missing events and events after October 26, 2026 at 00:00 UTC are right-censored at that cutoff; events exactly at the cutoff are observed. Survival means neither accepted-state change nor cancellation has occurred.

3.5 How did we verify the data?

3.5.1 Where is the prescription upload data coming from?

The report's upload analyses use `EDLDB.BT_HCA_HCDM.rxp_upload_entity`. Table 17 compares the entries in that source with `rxp_notification_service.upload_entity`, resolved in the `EDLDB` database, by `associated_resource_type`.

This verification covers all rows currently available in each source, without date, CMPU, vet-diet, upload-type, or study-population filters. In Table 17, each physical row counts once; no deduplication or joins are applied. These source counts therefore describe the source tables rather than the report's study cohort. They do not establish record-level equivalence between the two sources or document their upstream lineage.

3.5.1.1 How many upload-source entries have prescription IDs for each associated resource type?

Upload-source entries and entries with a non-zero prescription ID, by associated resource type.

Table 17: Upload-source entries and entries with a non-zero prescription ID, by associated resource type.

	`EDLDB.BT_HCA_HCDM.rxp_upload_entity`		`rxp_notification_service.upload_entity`	
`associated_resource_type`	All entries	With Rx ID	All entries	With Rx ID
DIET_PRESCRIPTION	181,943	181,943	184,637	184,635
DRUG_PRESCRIPTION	45,458	45,458	45,551	45,551
[SQL NULL]	1	0	41,475	0
Total	227,402	227,401	271,663	230,186

Extraction: 2026-11-02 16:02:32 UTC.

Notes: Rows include every resource type present in either source. A zero means that the source has no entries for that type. `[SQL NULL]` denotes a missing resource type, separately from any literal string value. Totals are queried independently from each source and reconciled against the sum of its grouped counts before publication. The extraction time above records the snapshot used for this comparison; rerunning the query may change the counts as sources are updated. *With Rx ID* counts entries with at least one populated, non-zero prescription identifier in HCDM `RX_ID` or notification-service `ASSOCIATED_RESOURCE_IDS`. Braces, array delimiters and quotes are removed before checking individual comma-separated IDs; SQL NULL, JSON null, blank, empty-list and numeric-zero values are excluded. An entry counts once regardless of how many IDs it contains. IDs are checked for presence, without joining to the prescription registry. Both count columns reconcile to their separately queried source totals. This table is refreshed independently of the subsequent order analyses; each analysis reports its own extraction time.

Table 18 shows the minimum and maximum creation timestamps of the entries currently available in each source, by resource type and overall. All timestamps are shown in UTC.

3.5.1.2 What creation-date ranges do the upload sources cover?

Creation-time coverage of current upload-source entries (UTC).

Table 18: Creation-time coverage of current upload-source entries (UTC).

	`EDLDB.BT_HCA_HCDM.rxp_upload_entity`		`rxp_notification_service.upload_entity`	
Resource type	Minimum created	Maximum created	Minimum created	Maximum created
DIET_PRESCRIPTION	2026-07-30 19:51:39	2026-11-04 04:36:23	2026-07-29 17:23:29	2026-11-04 13:20:17
DRUG_PRESCRIPTION	2026-10-01 14:46:41	2026-11-03 20:49:10	2026-10-01 14:46:41	2026-11-03 20:49:10
[SQL NULL]	2026-11-04 04:45:37	2026-11-04 04:45:37	2026-07-30 19:49:25	2026-11-04 13:21:37
Overall	2026-07-30 19:51:39	2026-11-04 04:45:37	2026-07-29 17:23:29	2026-11-04 13:21:37

Notes: No date, order-ID or prescription-ID filters are applied. HCDM uses `CREATED_TS_UTC`; notification-service `CREATED_AT` is converted to UTC before aggregation. Minimum and maximum ignore missing timestamps. Displayed values are rounded to the nearest second (half seconds round upward); the retained extract preserves full precision. Missing creation timestamps: 0 in HCDM and 0 in the notification-service source. The ranges describe available entries, without establishing continuous coverage or the full history of either system. Extraction: 2026-11-02 16:09:54 UTC.

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3.5.2 What's the distribution of all the vet diet prescriptions in the prescription upload tables by APPROVAL_CHANNEL ?

Table 19 joins all distinct prescription IDs in each upload source to EDLDB.BT_HCA_HCDM.RXP_PRESCRIPTIONS on rx_id, then retains matched prescriptions with IS_VET_DIET_FLAG = TRUE. Approval channels are taken from the prescription table. Each prescription counts once per source, even when it appears in multiple uploads. No date, CMPU, approval-channel, status or upload-resource-type filters are applied.

3.5.2.1 How are approval channels distributed among distinct uploaded vet-diet prescriptions?

Approval channels of distinct vet-diet prescriptions found in the upload sources.

Table 19: Approval channels of distinct vet-diet prescriptions found in the upload sources.

Approval channel	EDLDB.BT_HCA_HCDM.rxp_upload_entity Prescriptions (%)	rxp_notification_service.upload_entity Prescriptions (%)	Difference Notification minus HCDM
CUSTOMER_UPLOAD	157,922 (84.06%)	157,944 (84.05%)	+22
PH	21,232 (11.30%)	21,243 (11.31%)	+11
FAX	2,876 (1.53%)	2,877 (1.53%)	+1
CLINIC_EMAIL	2,187 (1.16%)	2,188 (1.16%)	+1
VERBAL	1,854 (0.99%)	1,854 (0.99%)	0
CUSTOMER_EMAIL	1,425 (0.76%)	1,425 (0.76%)	0
PARTNER	335 (0.18%)	335 (0.18%)	0
RHAPSODY	31 (0.02%)	31 (0.02%)	0
SNAIL_MAIL	9 (0.00%)	9 (0.00%)	0
RE_AUTH	1 (0.00%)	1 (0.00%)	0
Total	187,872 (100.00%)	187,907 (100.00%)	+35

Notes: Prescription IDs are expanded from HCDM RX_ID and notification-service ASSOCIATED_RESOURCE_IDS; missing, blank and zero tokens are excluded before joining. Vet-diet classification comes from the prescription registry, not the upload resource type. Percentages use the matched vet-diet total in each source. The difference is the notification-service count minus the HCDM count. The sources contain 232,111 and 233,285 distinct populated prescription IDs, respectively. Excluded non-vet-diet matches: 49,865 and 50,261; unmatched IDs: 0 and 677; matched IDs with unknown vet-diet flag: 0 and 0. Unmatched IDs cannot be assigned a vet-diet category or approval channel from this registry. NULL approval channels remain in the distribution when present. Both channel counts and join-coverage counts reconcile to independent Snowflake aggregates. Percentages may not sum to 100% because of rounding. Extraction: 2026-11-02 16:23:12 UTC.

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3.5.3 What is the coverage of vet diet prescriptions with APPROVAL_CHANNEL = CUSTOMER_UPLOAD, by RX_STATUS?

Table 20 starts from all vet-diet prescriptions in EDLDB.BT_HCA_HCDM.RXP_PRESCRIPTIONS with IS_VET_DIET_FLAG = TRUE and APPROVAL_CHANNEL = CUSTOMER_UPLOAD, created on or after July 30, 2026, 00:00 UTC, and shows upload-source coverage by PRESCRIPTION_SOURCE and current RX_STATUS. All statuses and prescription sources are included; there is no upper creation-date bound. Each qualifying prescription counts once.

3.5.3.1 How does upload-source coverage vary by prescription source and status for customer-upload vet-diet prescriptions?

Upload-source coverage by prescription source and current status for vet-diet customer-upload prescriptions created on or after July 30, 2026.

Table 20: Upload-source coverage by prescription source and current status for vet-diet customer-upload prescriptions created on or after July 30, 2026.

Prescription source	RX_STATUS	Eligible	HCDM (%)	Notification (%)	Both (%)	HCDM only	Notification only	Neither (%)
CMPU	SYSTEM_CLOSED	111,789	9,707 (8.68%)	9,707 (8.68%)	9,707 (8.68%)	0	0	102,082 (91.32%)
CMPU	APPROVED	50,723	32,394 (63.86%)	32,394 (63.86%)	32,394 (63.86%)	0	0	18,329 (36.14%)
CMPU	VET_REVIEW	7,897	352 (4.46%)	370 (4.69%)	352 (4.46%)	0	18	7,527 (95.31%)
CMPU	CLOSED	2,665	1,323 (49.64%)	1,323 (49.64%)	1,323 (49.64%)	0	0	1,342 (50.36%)
CMPU	DENIED	1	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	1 (100.00%)
CMPU	All statuses	173,075	43,776 (25.29%)	43,794 (25.30%)	43,776 (25.29%)	0	18	129,281 (74.70%)
CUSTOMER_UPLOAD	APPROVED	101,783	58,884 (57.85%)	58,884 (57.85%)	58,884 (57.85%)	0	0	42,899 (42.15%)
CUSTOMER_UPLOAD	SYSTEM_CLOSED	13,900	13,697 (98.54%)	13,697 (98.54%)	13,697 (98.54%)	0	0	203 (1.46%)
CUSTOMER_UPLOAD	CLOSED	4,387	1,938 (44.18%)	1,938 (44.18%)	1,938 (44.18%)	0	0	2,449 (55.82%)
CUSTOMER_UPLOAD	VET_REVIEW	1,404	895 (63.75%)	899 (64.03%)	895 (63.75%)	0	4	505 (35.97%)
CUSTOMER_UPLOAD	All statuses	121,474	75,414 (62.08%)	75,418 (62.09%)	75,414 (62.08%)	0	4	46,056 (37.91%)
PRACTICE_HUB	APPROVED	34,450	27,352 (79.40%)	27,352 (79.40%)	27,352 (79.40%)	0	0	7,098 (20.60%)
PRACTICE_HUB	CLOSED	1,998	1,377 (68.92%)	1,377 (68.92%)	1,377 (68.92%)	0	0	621 (31.08%)
PRACTICE_HUB	SYSTEM_CLOSED	771	40 (5.19%)	40 (5.19%)	40 (5.19%)	0	0	731 (94.81%)
PRACTICE_HUB	VET_REVIEW	3	3 (100.00%)	3 (100.00%)	3 (100.00%)	0	0	0 (0.00%)
PRACTICE_HUB	All statuses	37,222	28,772 (77.30%)	28,772 (77.30%)	28,772 (77.30%)	0	0	8,450 (22.70%)
FAX	APPROVED	5,813	4,379 (75.33%)	4,379 (75.33%)	4,379 (75.33%)	0	0	1,434 (24.67%)
FAX	CLOSED	457	312 (68.27%)	312 (68.27%)	312 (68.27%)	0	0	145 (31.73%)
FAX	SYSTEM_CLOSED	380	35 (9.21%)	35 (9.21%)	35 (9.21%)	0	0	345 (90.79%)
FAX	DENIED	4	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	4 (100.00%)
FAX	VET_REVIEW	4	1 (25.00%)	1 (25.00%)	1 (25.00%)	0	0	3 (75.00%)
FAX	All statuses	6,658	4,727 (71.00%)	4,727 (71.00%)	4,727 (71.00%)	0	0	1,931 (29.00%)
CLINIC_EMAIL	APPROVED	5,788	4,363 (75.38%)	4,363 (75.38%)	4,363 (75.38%)	0	0	1,425 (24.62%)
CLINIC_EMAIL	CLOSED	483	312 (64.60%)	312 (64.60%)	312 (64.60%)	0	0	171 (35.40%)
CLINIC_EMAIL	SYSTEM_CLOSED	282	19 (6.74%)	19 (6.74%)	19 (6.74%)	0	0	263 (93.26%)
CLINIC_EMAIL	VET_REVIEW	4	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	4 (100.00%)
CLINIC_EMAIL	DENIED	1	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	1 (100.00%)
CLINIC_EMAIL	All statuses	6,558	4,694 (71.58%)	4,694 (71.58%)	4,694 (71.58%)	0	0	1,864 (28.42%)
PARTNER_API	APPROVED	429	343 (79.95%)	343 (79.95%)	343 (79.95%)	0	0	86 (20.05%)
PARTNER_API	CLOSED	20	12 (60.00%)	12 (60.00%)	12 (60.00%)	0	0	8 (40.00%)
PARTNER_API	SYSTEM_CLOSED	10	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	10 (100.00%)
PARTNER_API	All statuses	459	355 (77.34%)	355 (77.34%)	355 (77.34%)	0	0	104 (22.66%)
DIGITAL	SYSTEM_CLOSED	143	31 (21.68%)	31 (21.68%)	31 (21.68%)	0	0	112 (78.32%)
DIGITAL	APPROVED	133	100 (75.19%)	100 (75.19%)	100 (75.19%)	0	0	33 (24.81%)
DIGITAL	CLOSED	3	2 (66.67%)	2 (66.67%)	2 (66.67%)	0	0	1 (33.33%)
DIGITAL	VET_REVIEW	3	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	3 (100.00%)
DIGITAL	All statuses	282	133 (47.16%)	133 (47.16%)	133 (47.16%)	0	0	149 (52.84%)
CVC	APPROVED	51	42 (82.35%)	42 (82.35%)	42 (82.35%)	0	0	9 (17.65%)
CVC	CLOSED	1	0 (0.00%)	0 (0.00%)	0 (0.00%)	0	0	1 (100.00%)
CVC	VET_REVIEW	1	1 (100.00%)	1 (100.00%)	1 (100.00%)	0	0	0 (0.00%)
CVC	All statuses	53	43 (81.13%)	43 (81.13%)	43 (81.13%)	0	0	10 (18.87%)
RHAPSODY	APPROVED	7	6 (85.71%)	6 (85.71%)	6 (85.71%)	0	0	1 (14.29%)
RHAPSODY	CLOSED	1	1 (100.00%)	1 (100.00%)	1 (100.00%)	0	0	0 (0.00%)
RHAPSODY	All statuses	8	7 (87.50%)	7 (87.50%)	7 (87.50%)	0	0	1 (12.50%)
All sources	All statuses	345,789	157,921 (45.67%)	157,943 (45.68%)	157,921 (45.67%)	0	22	187,846 (54.32%)

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Notes: Coverage requires at least one current upload-source entry with the same `rx_id`. HCDM `RX_ID` and notification-service `ASSOCIATED_RESOURCE_IDS` are split into individual trimmed IDs; NULL, blank and zero tokens are excluded. Each upload source is reduced to a distinct set of prescription IDs before joining, so multiple uploads do not increase the prescription count. The join uses `rx_id` only, without customer, order, upload-date or associated-resource-type restrictions. HCDM and Notification count any match in that source and overlap. Within each prescription-source/status group, Both, HCDM only, Notification only and Neither partition the eligible prescriptions. Percentages use the eligible count on that row. All statuses rows are prescription-source subtotals; All sources / All statuses is the overall total. Do not add subtotals to their detail rows. The two only columns show counts. SQL NULL source or status is retained when present. Percentages may not sum to 100% after rounding. No current match does not establish that an upload never existed. The row-level joins and independent Snowflake aggregates agree for every source/status group, source subtotal, status subtotal and overall. Extraction: 2026-11-02 17:49:55 UTC.

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3.5.4 What is the distribution of order cancellation reasons in the prescription upload tables?

This comparison includes entries with `associated_resource_type = DIET_PRESCRIPTION` and a non-null associated order ID in each source. HCDM uses `ORDER_ID`; the notification-service source uses `context:orderId`. Cancellation reasons are joined from `EDLDB.ECOM.ORDERS`. No date, CMPU, study-cohort, order-item, upload-type, or replication filters are applied. In HCDM, 137,431 of 176,401 diet-upload entries have an order ID, representing 115,929 distinct orders. In the notification-service source, 140,104 of 179,177 entries have an order ID, representing 116,223 distinct orders. Table 21 compares the cancellation reasons side by side. Each order counts once within each source regardless of the number of uploads.

3.5.4.1 How are cancellation reasons distributed among orders associated with vet-diet uploads?

Order cancellation reasons for diet-prescription uploads with associated order IDs.

Table 21: Order cancellation reasons for diet-prescription uploads with associated order IDs.

Order cancellation reason	EDLDB.BT_HCA_HCDM.rxp_upload_entity Orders (%)	rxp_notification_service.upload_entity Orders (%)	Difference Notification minus HCDM
No recorded reason (SQL NULL)	105,843 (88.62%)	105,940 (88.58%)	+97
Customer Initiated Cancellation	5,007 (4.19%)	5,010 (4.19%)	+3
Customer Request	4,462 (3.74%)	4,463 (3.73%)	+1
Aged Order	2,020 (1.69%)	2,032 (1.70%)	+12
Pharmacy Rejection	1,108 (0.93%)	1,108 (0.93%)	0
Prescription Issue	568 (0.48%)	569 (0.48%)	+1
Other	274 (0.23%)	274 (0.23%)	0
Duplicate Order	57 (0.05%)	57 (0.05%)	0
No Response	48 (0.04%)	48 (0.04%)	0
Payment Declined	29 (0.02%)	29 (0.02%)	0
No matching ECOM order	0 (0.00%)	53 (0.04%)	+53
Fraud	14 (0.01%)	14 (0.01%)	0
Payment Issue	2 (0.00%)	2 (0.00%)	0
Reseller	1 (0.00%)	1 (0.00%)	0
Total	119,433 (100.00%)	119,600 (100.00%)	+167

Notes: Percentages use each source's distinct-order total, and the difference is the notification-service order count minus the HCDM order count. Totals include unmatched IDs; the two source populations are not restricted to their intersection. 115,929 order IDs occur in both sources, 0 occur only in HCDM and 294 occur only in the notification-service source. Unmatched order IDs: 0 in HCDM and 143 in the notification-service source. A NULL reason does not imply that an order was not canceled. Of 13,266 HCDM-linked orders and 13,277 notification-linked orders with status `Canceled`, 134 and 134, respectively, have a NULL reason. Percentages may not sum to 100% because of rounding. Upload extraction: 2026-11-02 15:45:52.376 Z; order extraction: 2026-11-02 15:45:54.406 Z (UTC).

3.5.5 What are the order cancellation reasons for null associated resource type entries in the rxp_notification_service.upload_entity table?

Of the 40,451 notification-service upload entries with `associated_resource_type IS NULL`, 1,998 (4.94%) have both a non-null `context:orderId` and `context:orderItemId`; 38,453 have neither. These entries reference 1,726 distinct orders, of which 1,724 match `EDLDB.ECOM.ORDERS` and 2 are unmatched. Table 22 shows the cancellation-reason distribution across these distinct order IDs.

3.5.5.1 How are cancellation reasons distributed among orders referenced by uploads with no associated resource type?

Cancellation reasons for orders referenced by uploads with no associated resource type.

Table 22: Cancellation reasons for orders referenced by uploads with no associated resource type.

Order cancellation reason	Distinct orders	Share (%)
No recorded reason (SQL NULL)	1,644	91.69
Aged Order	74	4.13
Customer Initiated Cancellation	45	2.51
Customer Request	15	0.84
Prescription Issue	8	0.45
Pharmacy Rejection	4	0.22
Fraud	1	0.06
Other	1	0.06
No matching ECOM order	1	0.06
Total	1,793	100.00

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Notes: All available NULL-type uploads are included, without date or other filters. Missing JSON keys and JSON null count as missing identifiers. Each order counts once regardless of upload count; percentages use all 1,726 distinct order IDs, including unmatched IDs. A NULL cancellation reason does not imply that the order was not canceled. Of 139 matched orders with status `Canceled`, 2 have a NULL reason and 67 (48.20%) have reason `Aged Order`. Percentages may not sum to 100% because of rounding. Upload extraction: 2026-11-02 15:35:14.335 Z; order extraction: 2026-11-02 15:35:15.577 Z (UTC).

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3.5.6 How are data sources connected?

The diagrams describe the analytical relationships used throughout this report, including the executive summary's underlying tables and figures. They do not assert database-enforced foreign keys or upstream replication lineage. All physical sources resolve in `EDLDB`; `HCDM` abbreviates `BT_HCA_HCDM`. Blue solid boxes are physical tables or views; amber dashed boxes are derived datasets. Arrows carry source evidence into a result; edge labels state join fields and direction, rather than implying cardinality. The two upload sources are compared independently, without a record-to-record join between them. The notification source is used in verification, while the later study figures use HCDM uploads.

3.5.6.1 How do the report's source tables and analyses connect?

Report-wide source connections; details and matching rules follow.

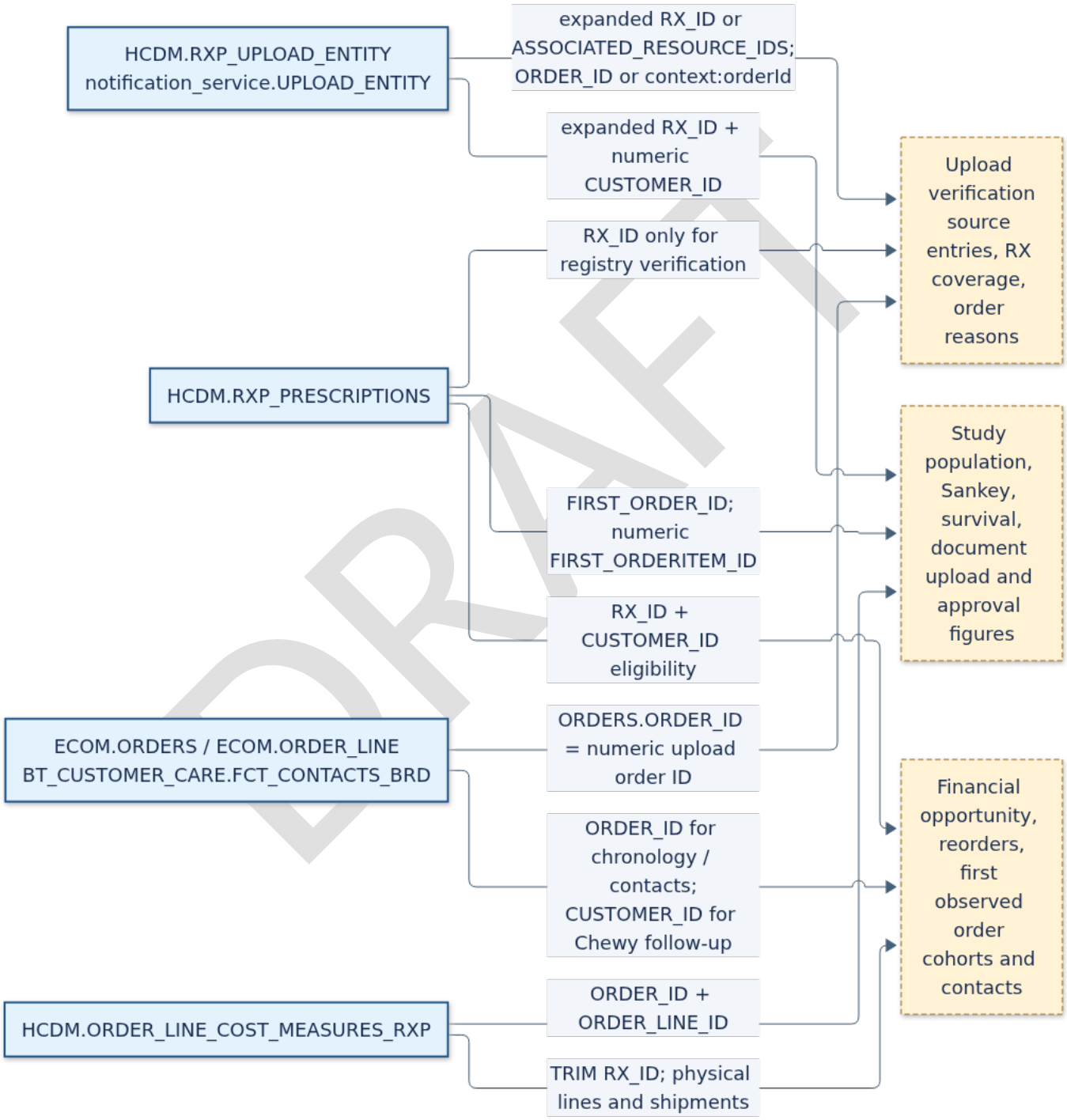


Figure 7: Report-wide source connections; details and matching rules follow.

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Source grains and counting units.

3.5.6.2 What is the grain and counting unit of each source?

The table below records the source grains and counting units used throughout the analysis.

Source (within EDLDB)	Grain used by this report
HCDM.RXP_UPLOAD_ENTITY	Physical upload row; entity analyses validate UPLOAD_ID and count it once, despite multiple expanded prescription IDs.
RXP_NOTIFICATION_SERVICE.UPLOAD_ENTITY	Physical upload row; prescription IDs may be a list. Context order and item IDs are separate fields.
HCDM.RXP_PRESCRIPTIONS	Prescription RX_ID; analyses check uniqueness or consistency of repeated registry values.
HCDM.ORDER_LINE_COST_MEASURES_RXP	Shipment × physical line: ORDER_ID, ORDER_LINE_ID, SHIPMENT_KEY. A line can have multiple shipments.
ECOM.ORDERS	Order header ORDER_ID; linked-order verification checks uniqueness, chronology analyses check consistent repeated values.
ECOM.ORDER_LINE	Physical line ORDER_ID, ORDER_LINE_ID; NET_SALES is already the full line amount.
BT_CUSTOMER_CARE.FCT_CONTACTS_BRD	Contact leg/session evidence, identified by order, SESSION_ID and LEG_ID for conflict checks; published counts use distinct customers, orders or relative days.

Duplicates and aggregation. Source-entry counts and timestamp extrema use all physical upload rows without deduplication. Prescription verification expands IDs and reduces them to distinct source/prescription sets, preventing upload multiplicity from multiplying registry counts. Coverage preserves eligible prescriptions with left joins; the channel distribution excludes unmatched and non-vet-diet IDs from its denominator. Order-reason comparisons count each normalized order once per source, retaining unmatched IDs separately from matched orders whose reason is SQL NULL. `context:orderItemId` is audited for presence in the NULL-resource analysis; it is not a join key for those order-level results.

For first-order figures, shipment rows collapse to one `(ORDER_ID, ORDER_LINE_ID)` before prescription association. Attribute-variant checks prevent conflicting shipment evidence from being silently resolved by SQL `MIN/MAX`. Multiple prescription associations are retained for audit, then distinct prescriptions, orders or lines are counted according to each output's grain. Missing or invalid first-order keys are audited outside the matched-line denominator. The decision extract uses an inner join because it describes matched first-order populations only.

Financial calculations instead retain one amount per distinct `(ORDER_ID, ORDER_LINE_ID, SHIPMENT_KEY)` and sum shipments to lines; identical amounts on different shipments remain separate contributions. Repeated associations cannot multiply sales. The baseline includes only shipment or cancellation records in its reporting window. Chewy-wide follow-up uses `ECOM.ORDER_LINE.NET_SALES` once per line, without summing it as shipment data. Consistent duplicates may collapse; conflicting key values fail the relevant generator's validation. Totals count distinct customers and orders again, rather than adding overlapping category counts.

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Upload verification joins. The channel and coverage tables match normalized prescription ID only, with no customer, upload-date or order restriction. The coverage cohort applies its registry creation cutoff and vet-diet/customer-upload filters; the channel comparison covers all uploaded vet-diet prescriptions. Numeric order IDs require trimmed digits before conversion. Source-order extracts are merged in Python with validated ECOM order headers using a left join.

3.5.6.3 How are upload IDs expanded and matched to prescriptions and orders?

Upload-source prescription expansion, registry matching and context-order matching.

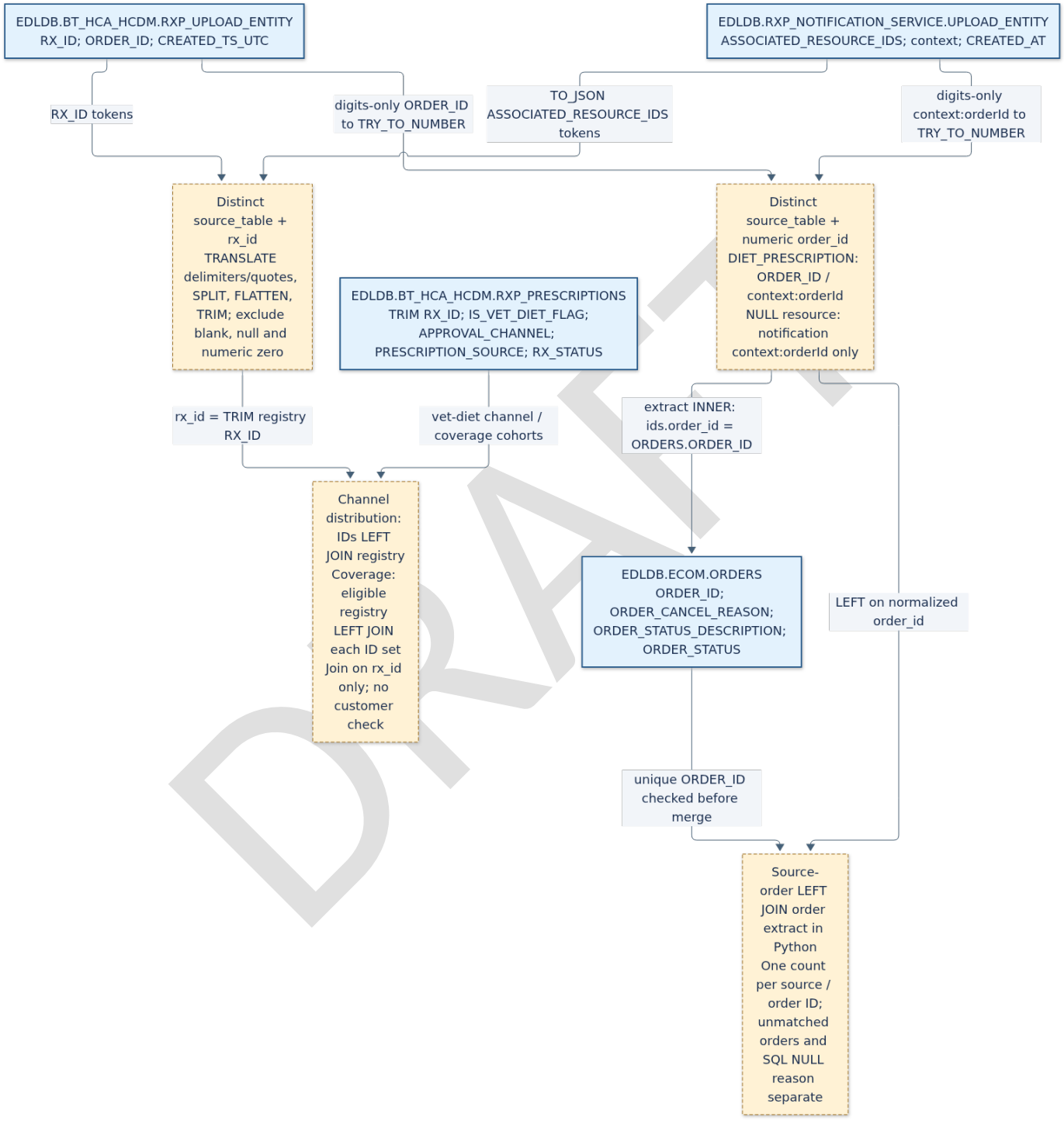


Figure 8: Upload-source prescription expansion, registry matching and context-order matching.

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Prescription, first-order and upload evidence. The study population uses CMPU vet-diet prescriptions created August 15 through October 14, 2026, UTC; focused figures also require `APPROVAL_CHANNEL = CUSTOMER_UPLOAD`. Both first-order keys are required: `FIRST_ORDER_ID = ORDER_ID` and the digits-only numeric conversion of `FIRST_ORDERITEM_ID = ORDER_LINE_ID`. Upload matching additionally requires registry `CUSTOMER_ID` to equal the numeric upload customer ID.

3.5.6.4 How are prescriptions matched to first-order lines and upload evidence?

First-order line matching and two distinct ways of selecting upload evidence.

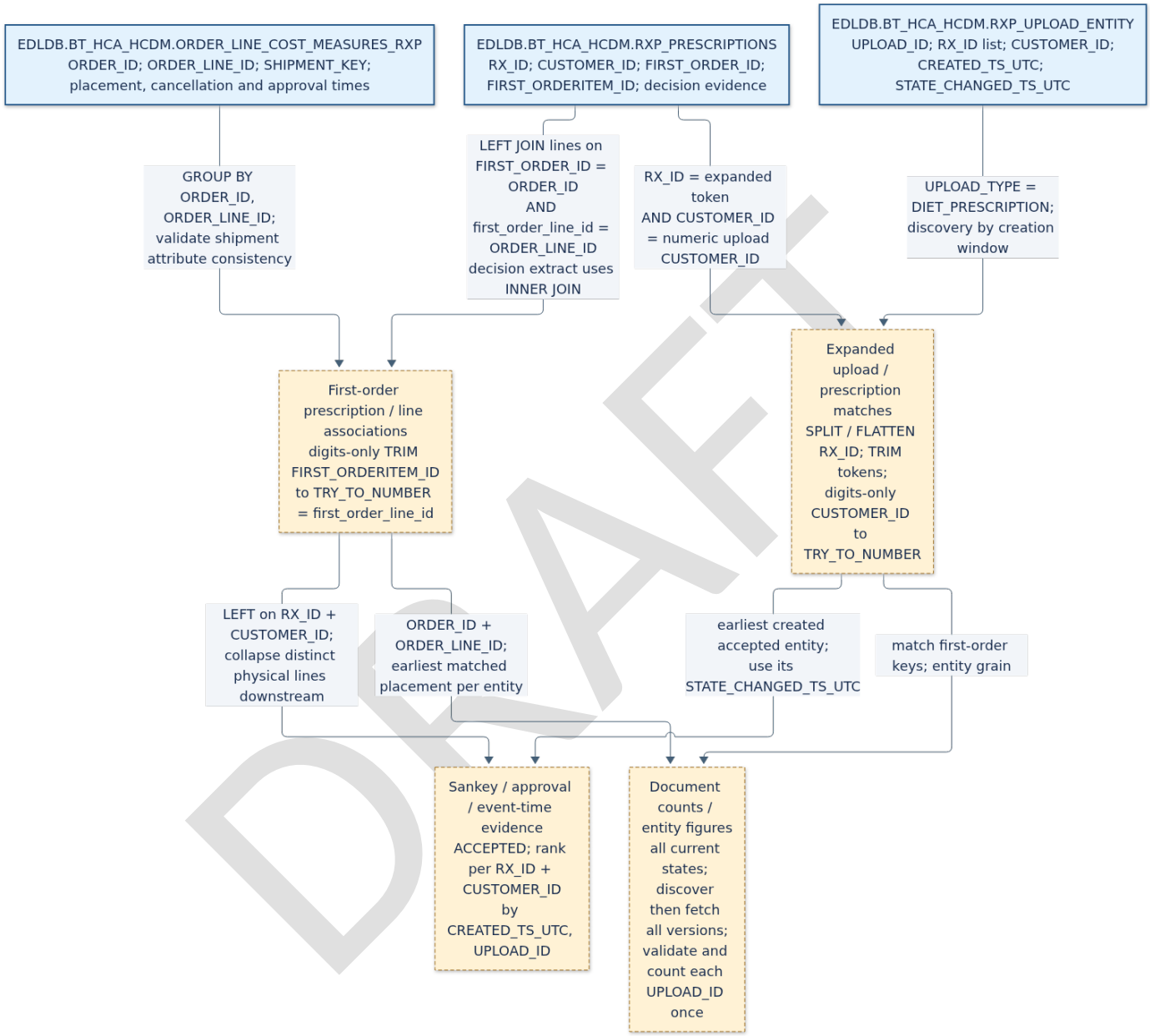


Figure 9: First-order line matching and two distinct ways of selecting upload evidence.

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Selection and observation windows. The Sankey, approval and survival pipelines select the earliest-created currently `ACCEPTED` upload per prescription/customer, breaking ties by `UPLOAD_ID`; the measured upload time is that entity's `STATE_CHANGED_TS_UTC`, not its creation time. Survival selects the earliest-created associated entity again at line grain. Approval comparisons use the line's `MAX_APPROVED_DATETIME`; survival uses `MIN_APPROVED_DENIED_DATETIME` for its 96% interval estimate when actual upload time is unavailable, then compares the candidate with cancellation. These estimates are derived events, not extra source tables.

Entity-state figures and document counts retain all current states, discover entities by creation time, fetch all versions of discovered IDs to expose conflicts, and count each entity once. The order-relative entity figure uses its earliest populated matched first-order placement. Upload discovery starts August 1; the Sankey/approval/entity figures use an exclusive October 26 bound, while survival uses an exclusive October 27 bound. Unknown creation times are retained for entity auditing. Current state changes and prescription decisions can occur after the discovery or aged-cancellation cutoff; these are snapshots, not historical event streams.

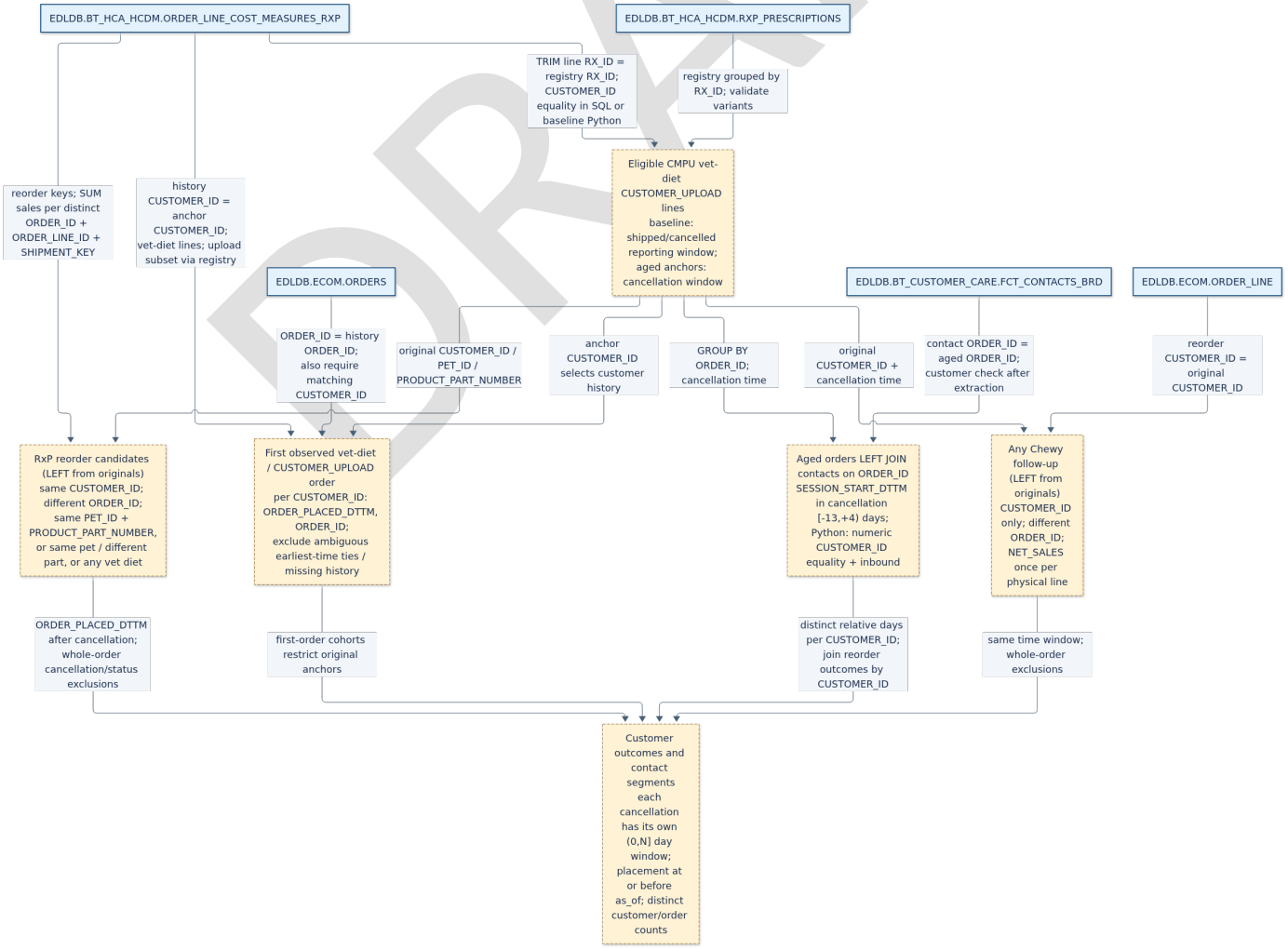
The study's first-versus-all-order comparison additionally matches `TRIM(line.RX_ID) = prescription.RX_ID` and customer ID for all linked orders, without an order-placement window. Its frozen physical keys are passed to `prescription_cohort_order_sales.sql` for shipment financial enrichment on both order and line ID. This differs from joining only `FIRST_ORDER_ID`. Decision evidence comes from the registry plus the matched line's approval/denial fields; no separate decision table is joined.

Financial, reorder and contact rules. The financial baseline's SQL left-joins the registry on trimmed line `RX_ID`; Python then requires nonmissing, equal numeric customer IDs and the CMPU/vet-diet/customer-upload flags. Aged-anchor queries apply these same customer and prescription checks directly in SQL. The baseline's shipped-or-cancelled window and the anchors' cancellation window are May 1 inclusive to August 1 exclusive, 2026; they do not impose the study's prescription-creation window.

First observed vet-diet and first customer-upload vet-diet cohorts use RxP line history, then ECOM order headers by order and customer ID. Chronology ranks by placement then order ID; earliest-time ties, missing placements and missing headers are audited and excluded from unambiguous first-order cohorts. Vet-diet history uses line vet-diet flags without requiring a registry match; customer-upload history also matches registry prescription/customer IDs and restricts that channel. Both histories span all prescription sources and dates. This is observed history, not a claim of complete lifetime coverage.

3.5.6.5 How are financial lines, later orders, and customer contacts connected?

Financial eligibility, order chronology, reorder matching and contact matching.



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Figure 10: Financial eligibility, order chronology, reorder matching and contact matching.

RxP follow-up matches exact customer, non-null pet and nonblank product part number for the same-product endpoint. Its other endpoints use the same pet with a different part, or any vet-diet line for that customer. Chewy-wide follow-up matches customer only in `ECOM.ORDER_LINE`. All endpoints require a different order placed strictly after its matched cancellation and through the specified follow-up day inclusive, bounded by the October 28, 2026, 00:00 UTC observation cutoff. Windows of 7–70 days overlap; each cancellation is its own anchor. Whole candidate orders with cancellation evidence or unknown line status are excluded. Orders and sales are deduplicated across repeated anchor matches; a returning customer counts once within each endpoint/window.

Contacts join the original aged order on `ORDER_ID` in SQL; Python normalizes identifiers and additionally requires equal nonmissing customer IDs, inbound direction and a populated timestamp. Contact legs are checked for conflicts and exact duplicate rows collapse. Relative days are elapsed UTC 24-hour bins, from $[-13, +4)$ days around cancellation, rather than calendar dates. The report includes handled, unhandled and unknown-handled inbound attempts. Frequency counts distinct relative days across a customer's aged orders. The contact-frequency and -7 through $+3$ contact-window comparisons join customer-level segments to verified 14-day same-pet/part outcomes by `CUSTOMER_ID`, retaining customers without a reorder in the denominator.

Table 6 uses the broader financial shipment/cancellation cohort and the same contact order/customer joins, grouping distinct contacting customers by current reason. No creation-relative day bounds apply; contacts are capped at the latest observed shipment/cancellation event per order and reason. Its SQL and generator are `src/sql/opportunity_customer_contacts_by_cancel_reason.sql` and `src/python/create_opportunity_contacts_by_cancel_reason.py`.

Figure 11 isolates the contact-matching pipeline, including the customer checks applied after the SQL extraction.

3.5.6.6 How are customer contacts matched to aged orders and observation windows?

Contact matching: aged-order eligibility, SQL order/time join, Python customer checks and distinct counting.

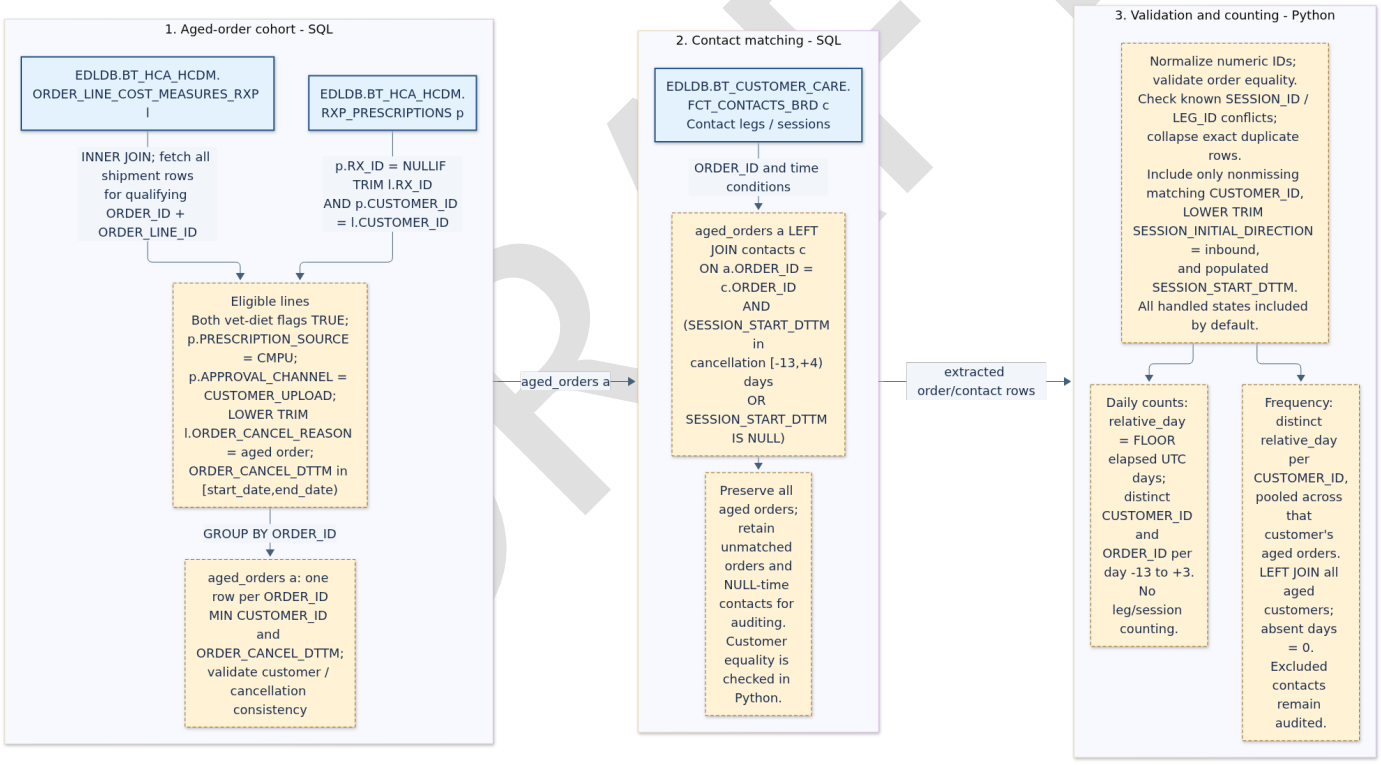


Figure 11: Contact matching: aged-order eligibility, SQL order/time join, Python customer checks and distinct counting.

Reproduction and source trace. Run `make data-source-diagrams` to generate the editable Mermaid sources and transparent PNGs; `make report` also rebuilds missing or outdated diagrams. The tracked definitions are in `src/python/create_data_source_diagrams.py`; the Markdown grain table is `outputs/tables/data-source-connections/source-grains.md`. The following implementation files define the relationships above; independently aggregated verification SQL checks the same populations without adding sources. Unused exploratory queries are excluded from this map.

- Verification: `src/sql/upload_data_verification.sql`, `upload_source_time_coverage.sql`, `upload_vet_diet_approval_channels.sql`, `cmpu_approved_prescription_upload_coverage.sql`, `diet_upload_source_orders.sql`, `diet_upload_linked_orders.sql`, `null_resource_upload_context.sql` and `null_resource_upload_orders.sql` (all under `src/sql/`); their corresponding `src/python/analyze_*.py` modules validate and merge the retained extracts.
- Study figures: `src/sql/cmpu_first_order_line_status.sql`, `cmpu_first_order_line_event_times.sql`, `cmpu_first_order_line_approval_distribution.sql`, `cmpu_upload_to_max_approval_extract.sql`, `cmpu_u`

pload_entity_state_change_distributions.sql, cmpu_upload_entity_order_creation_distributions.sql, cmpu_study_population_uploaded_documents.sql and cmpu_study_population_prescription_decisions.sql; the corresponding src/python/create_cmpu_*.py modules enforce line/entity grains. The frozen all-order membership query is retained under data/opportunity/prescription-cohort-orders/; financial enrichment uses src/sql/prescription_cohort_order_sales.sql.

- Financial and contacts: src/sql/opportunity_order_cancellation_summary.sql, opportunity_aged_reorders.sql, opportunity_new_upload_customer_follow_up.sql, opportunity_new_vet_diet_customer_follow_up.sql, and the three opportunity_*chewy_follow_up.sql templates; opportunity_aged_first_vet_diet_order_customers.sql, opportunity_aged_first_customer_upload_vet_diet_order_customers.sql and opportunity_aged_customer_contacts.sql define history and contact extracts. The corresponding opportunity Python generators aggregate outcomes; src/python/create_opportunity_contact_frequency_reorders.py combines contact segments and reorder outcomes. Retained rendered queries, source fingerprints and local verification SQL are under data/opportunity/.

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References

[1] *CMPU Aged Order Analysis*. GitHub repository. https://github.com/Chewy-Inc/phds-ds-discovery-cmpu-vet-diet-cancelled-order.

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